

DYNAMICS OF RIGID BODIES

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General equations of Motion

- The **kinetics** of rigid bodies treats the relationships between the external forces acting on a body and the corresponding translational and rotational motions of the body.
- **Plane Motion:** A body which can be approximated as a thin slab with its motion confined to the plane of the slab
- **Plane of Motion:** The plane of motion will contain the mass center, and all forces which act on the body will be projected onto the plane of motion
- A body which has appreciable dimensions normal to the plane of motion but is symmetrical about that plane of motion through the mass center may be treated as having plane motion.

General equations of Motion

- **Two force equations and one moment equation** or their equivalent are required to determine the state of rigid-body plane motion.
- Successful application of kinetics requires that you isolate the body or system to be analyzed
- For problems involving the instantaneous relationships among force, mass, and acceleration, the body or system should be explicitly defined by isolating it with its free-body diagram.
- When the principles of work and energy are employed, an active force diagram which shows only those external forces which do work on the system may be used in lieu of the free-body diagram.
- No solution of a problem should be attempted without first defining the complete external boundary of the body or system and identifying all external forces which act on it.

Mass Moment of Inertia

- **Mass moment of inertia of the body:** Property of the body which accounts for the radial distribution of its mass with respect to a particular axis of rotation normal to the plane of motion

General equations of Motion

Methods to obtain equations of motion

- Forces and moments to the instantaneous linear and angular accelerations
- Method of work and energy
- Method of impulse and momentum

Three types of motion:

- Translation
- fixed-axis rotation
- general plane motion

General equations of Motion

- Forces and moments to the instantaneous linear and angular accelerations
- Resultant $\sum F$ of the external forces acting on the body equals the mass m of the body times the acceleration of its mass center G .

$$\sum F = m\bar{a}$$

- Moment equation taken about the mass center, resultant moment about the mass center of the external forces on the body equals the time rate of change of the angular momentum of the body about the mass center.

$$\sum M_G = \dot{H}_G$$

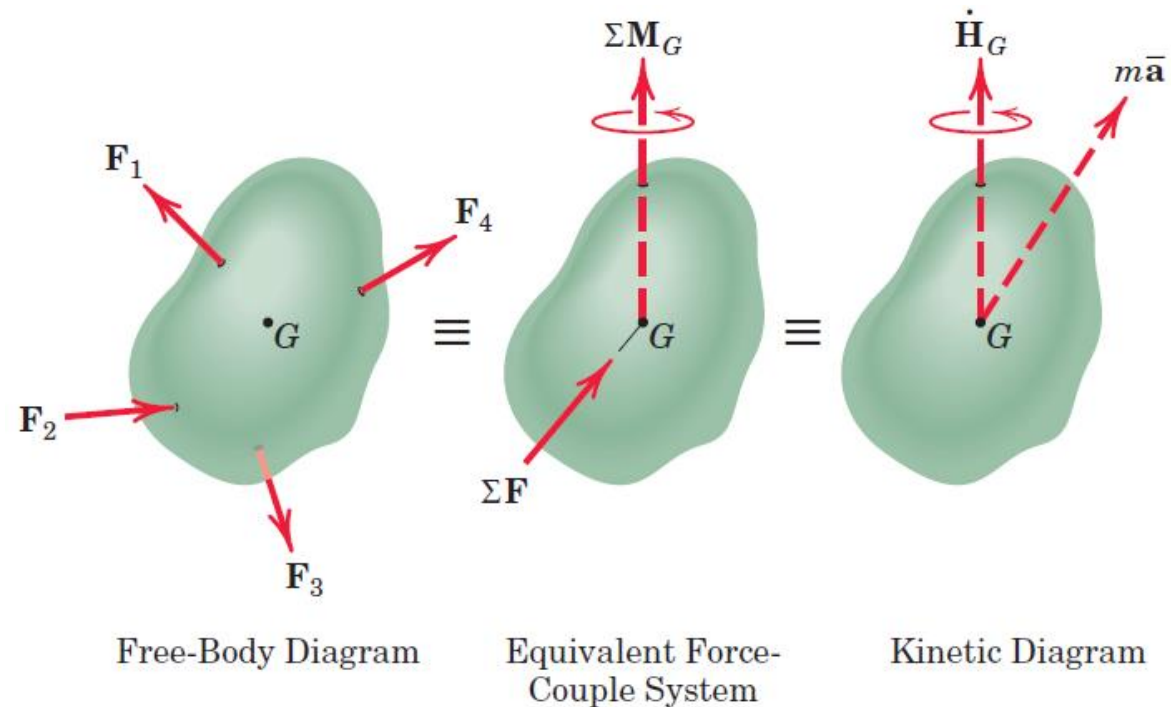
General equations of Motion

Free Body Diagram and Kinetic Diagram

- By replacing the external forces by their equivalent force-couple system in which the resultant force acts through the mass center, we may visualize the action of the forces and the corresponding dynamic response of the body with the aid of figure.

$$\sum F = m\bar{a}$$

$$\sum M_G = \dot{H}_G$$

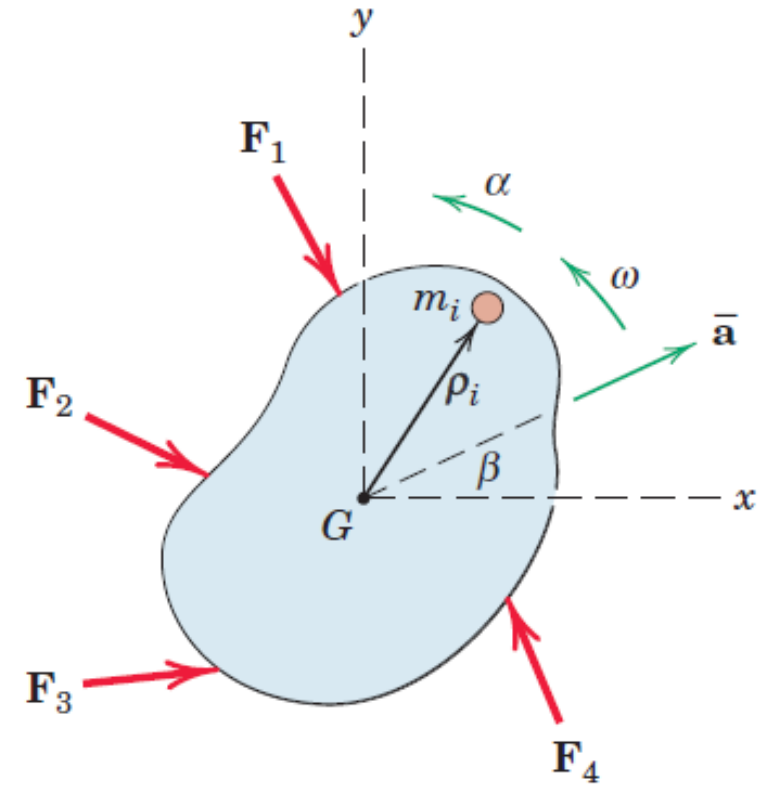


Plane Motion Equations

- Mass Center G
- Acceleration of mass center $\bar{\mathbf{a}}$
- Angular Velocity of body $\omega \hat{\mathbf{k}}$
- Angular acceleration of body $\alpha = \dot{\omega} \hat{\mathbf{k}}$
- Angular momentum about the mass center for the general system

$$\mathbf{H}_G = \sum \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i$$

$\boldsymbol{\rho}_i$ is position vector of m_i relative to G





Plane Motion Equations

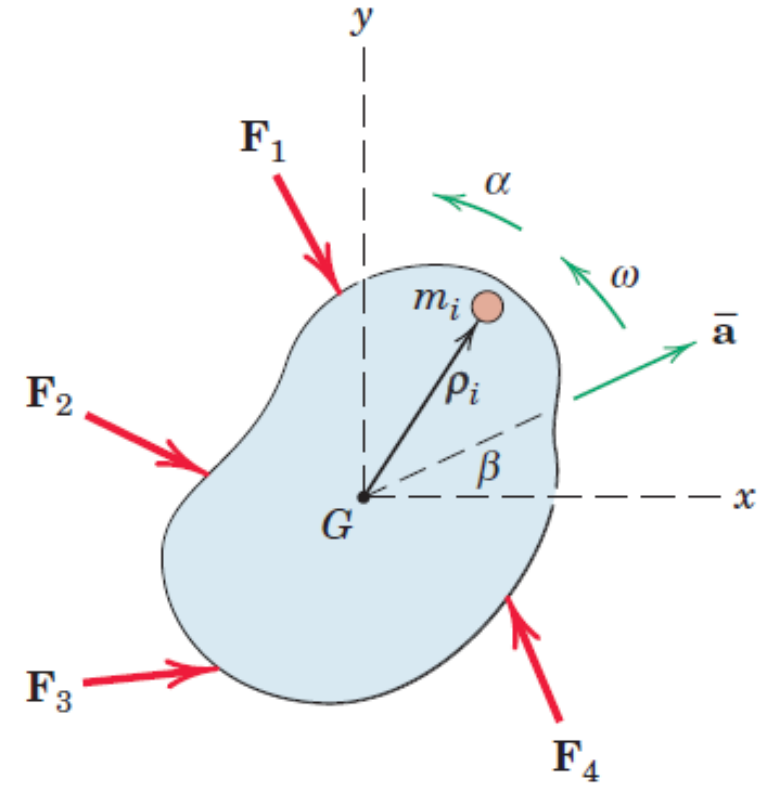
- Angular momentum about the mass center for the general system

$$\mathbf{H}_G = \sum \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i$$

$\boldsymbol{\rho}_i$ is position vector of m_i relative to G

- For rigid body, $\dot{\boldsymbol{\rho}}_i = \boldsymbol{\omega} \times \boldsymbol{\rho}_i$
- Direction of $\dot{\boldsymbol{\rho}}_i$ is perpendicular to $\boldsymbol{\rho}_i$ in x-y plane
- Direction of $\mathbf{H}_G$ is along $\boldsymbol{\omega}$
- Magnitude of $\mathbf{H}_G$ is $\sum \boldsymbol{\omega} m_i \rho_i^2 = \boldsymbol{\omega} \sum m_i \rho_i^2 = \boldsymbol{\omega} \int \rho^2 dm$
- **Mass moment of Inertia of the body about the z-axis through G , $\bar{I} = \int \rho^2 dm$**

$$\mathbf{H}_G = \boldsymbol{\omega} \int \rho^2 dm = \bar{I} \boldsymbol{\omega}$$









Plane Motion Equations

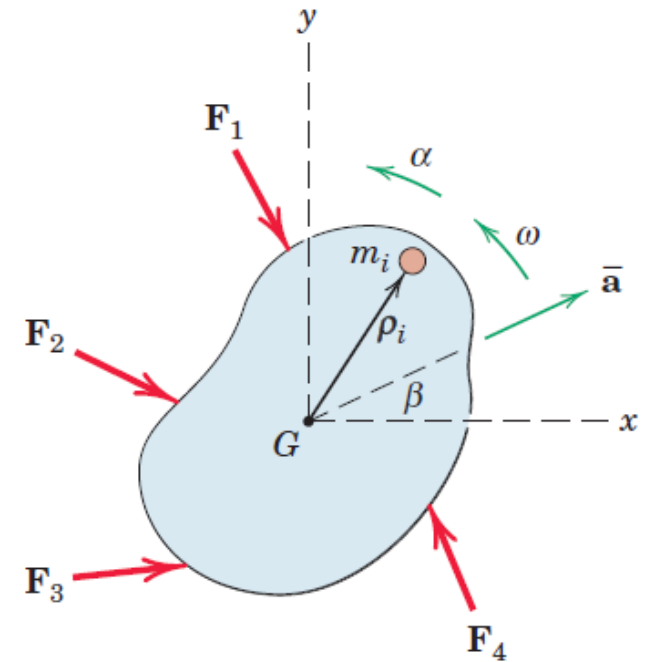
- Angular momentum of the rigid body about z-axis passing through G

$$H_G = \omega \int \rho^2 dm = \bar{I} \omega$$

- $\bar{I}$ is constant property of the body
- Measure of the rotational inertia, which is the resistance to change in rotational velocity due to the radial distribution of mass around the z-axis through G
- Moment Equation

$$M_G = \dot{H}_G = \bar{I} \dot{\omega} = \bar{I} \alpha$$

- $\dot{\omega} = \alpha$ is angular acceleration of the body



Plane Motion Equations

- Moment Equation

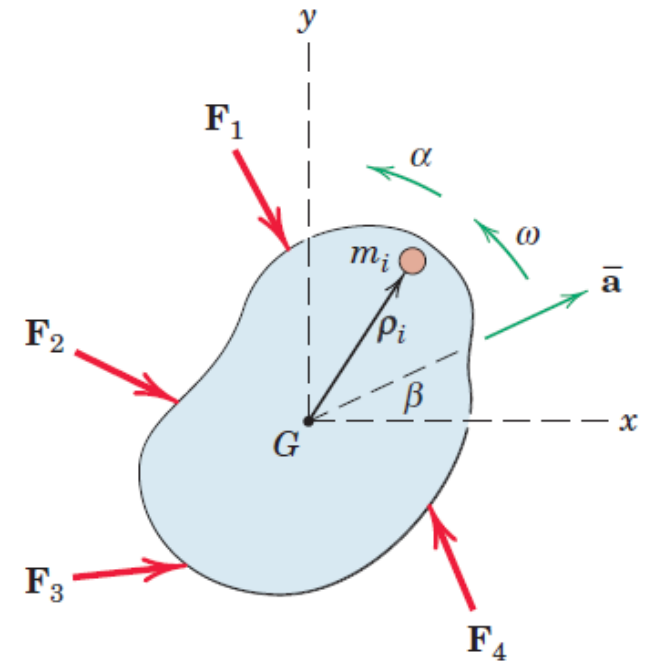
$$M_G = \bar{I}\alpha$$

- $\dot{\omega} = \alpha$ is angular acceleration of the body

- Force Equation

$$F = m\bar{a}$$

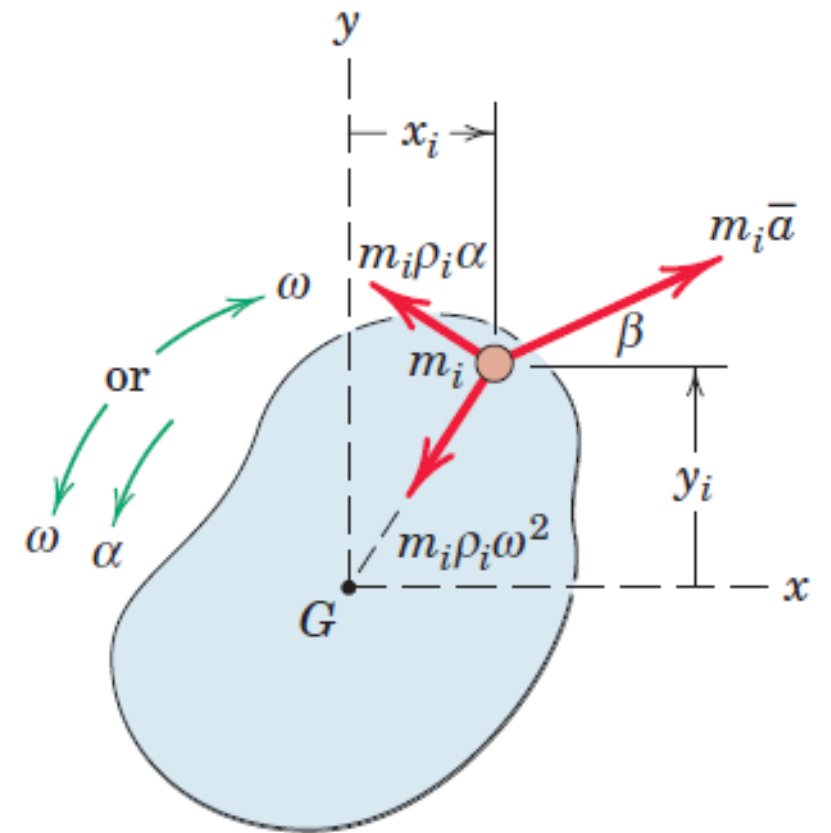
- $\bar{a}$ is acceleration of the mass center of body



Alternate derivations

- The acceleration of m_i equals the vector sum of $\bar{\mathbf{a}}$ and the relative terms $\rho_i \omega^2$ and $\rho_i \alpha$, where the mass center G is used as the reference point
- Resultant of all forces on m_i has the components $m_i \bar{\mathbf{a}}$, $m_i \rho_i \omega^2$ and $m_i \rho_i \alpha$ in the directions shown
- Sum of the moments of these force components about G in the sense of α
- $M_{G_i} = m_i \rho_i^2 \alpha + m_i \bar{\mathbf{a}} \sin \beta x_i + m_i \bar{\mathbf{a}} \cos \beta y_i$
- $\sum M_G = \sum m_i \rho_i^2 \alpha + \bar{\mathbf{a}} \sin \beta \sum m_i x_i + \bar{\mathbf{a}} \cos \beta \sum m_i y_i$

$$\sum m_i x_i = m \bar{x} = 0 \quad \sum m_i y_i = m \bar{y} = 0 \quad \sum M_G = \sum m_i \rho_i^2 \alpha = \bar{I} \alpha$$

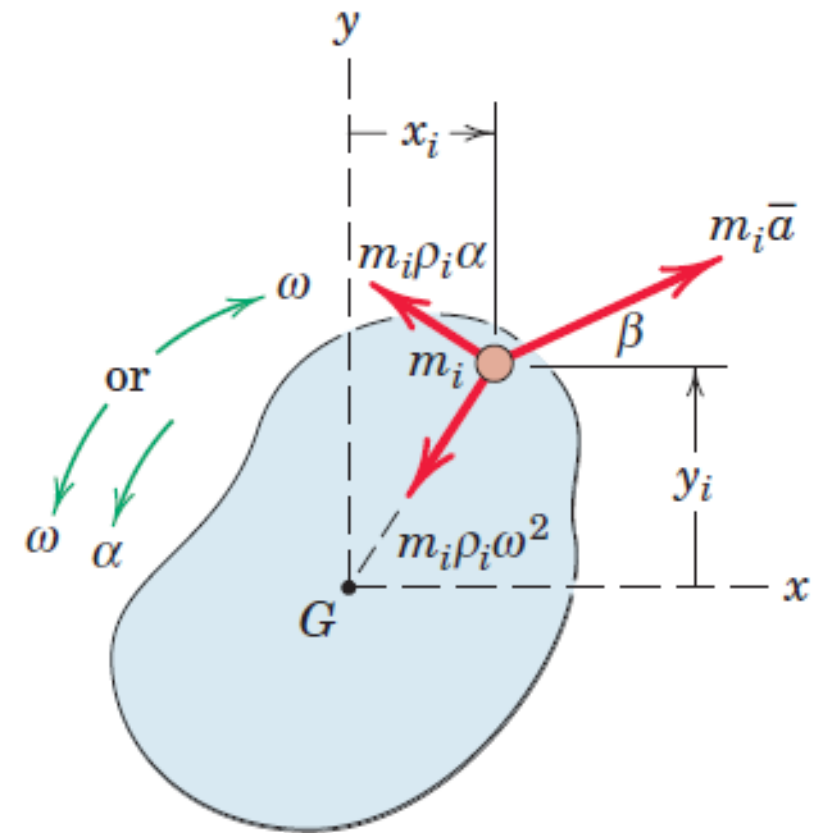


Alternate derivations

- The contribution to $\sum M_G$ of the forces internal to the body is **zero** since they occur in pairs of equal and opposite forces of action and reaction between interacting particles
- $\sum M_G$ represents the sum of moments about the mass center G of only the external forces acting on the body, as disclosed by the free-body diagram.

$$\sum M_G = \sum m_i \rho_i^2 \alpha = \bar{I} \alpha$$

- Force component $m_i \rho_i \omega^2$ has no moment about G
- The angular velocity ω has no influence on the moment equation about the mass center



Plane Motion Equations

➤ Moment Equation

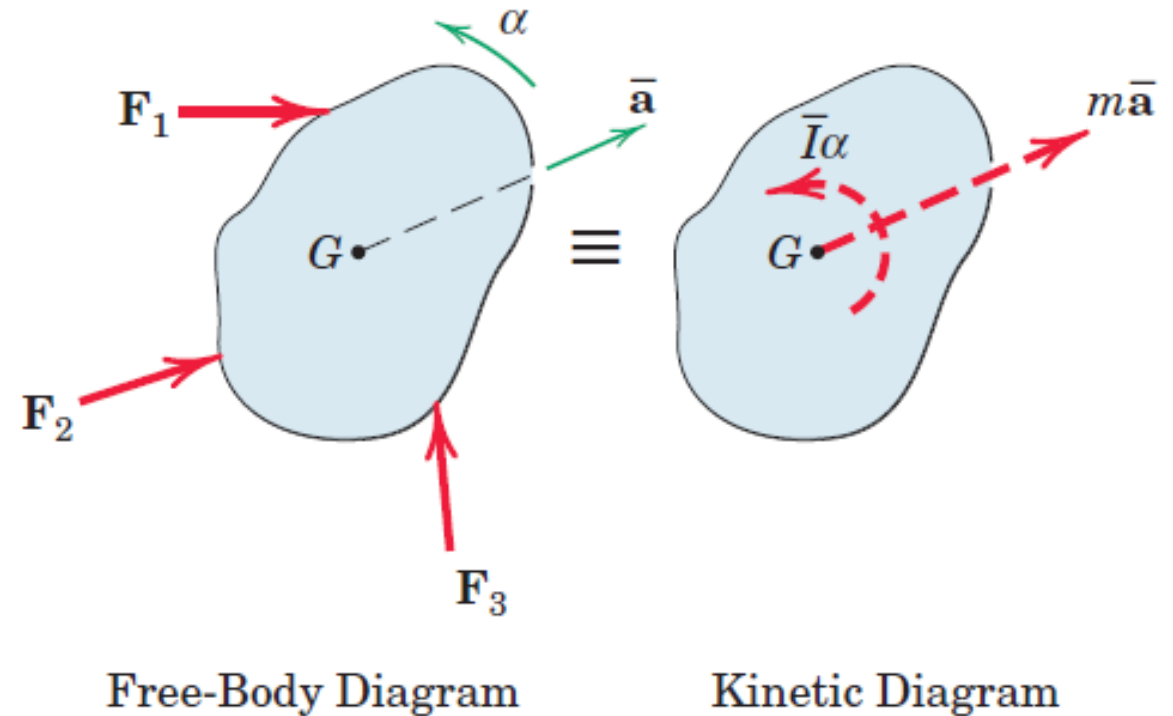
$$M_G = \bar{I}\alpha$$

➤ $\dot{\omega} = \alpha$ is angular acceleration of the body

➤ Force Equation

$$F = m\bar{a}$$

➤ $\bar{a}$ is acceleration of the mass center of body



Alternate derivations

- General equation for moments about an arbitrary point P

$$\sum M_p = \dot{H}_G + \bar{\rho} \times m\bar{a}$$

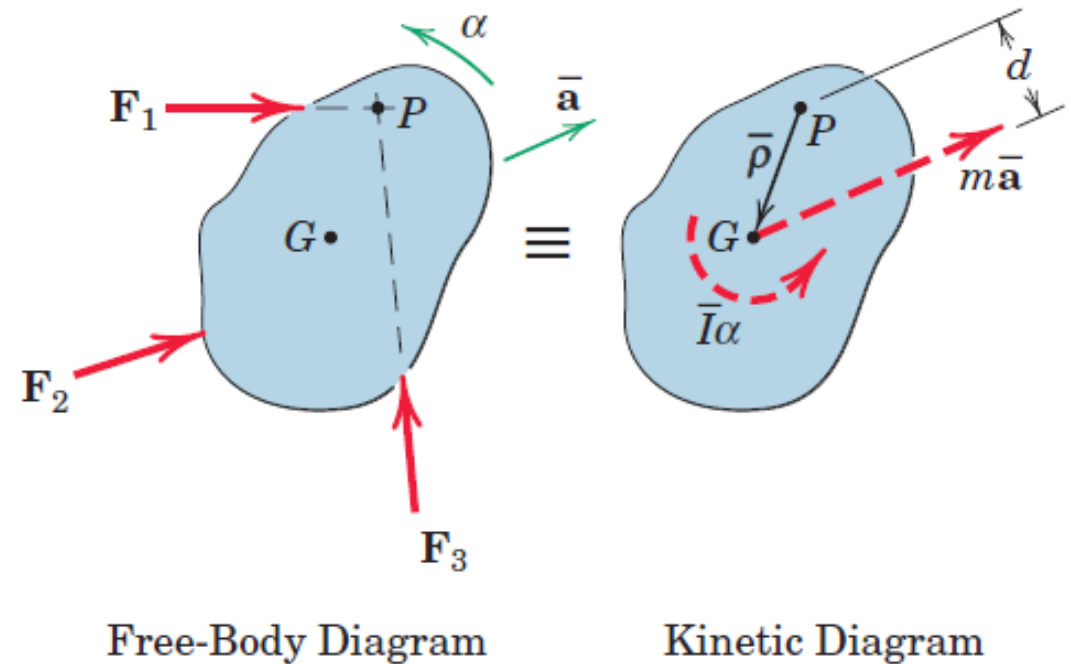
- $\bar{\rho}$ is a vector from point P to point G

- $\bar{a}$ is mass center acceleration

$$\dot{H}_G = \bar{I}\alpha \qquad \bar{\rho} \times m\bar{a} = m\bar{a}d$$

$$\sum M_p = \bar{I}\alpha + m\bar{a}d$$

All three terms are positive in the counterclockwise sense for the example shown, and the choice of P eliminates reference to F_1 and F_3





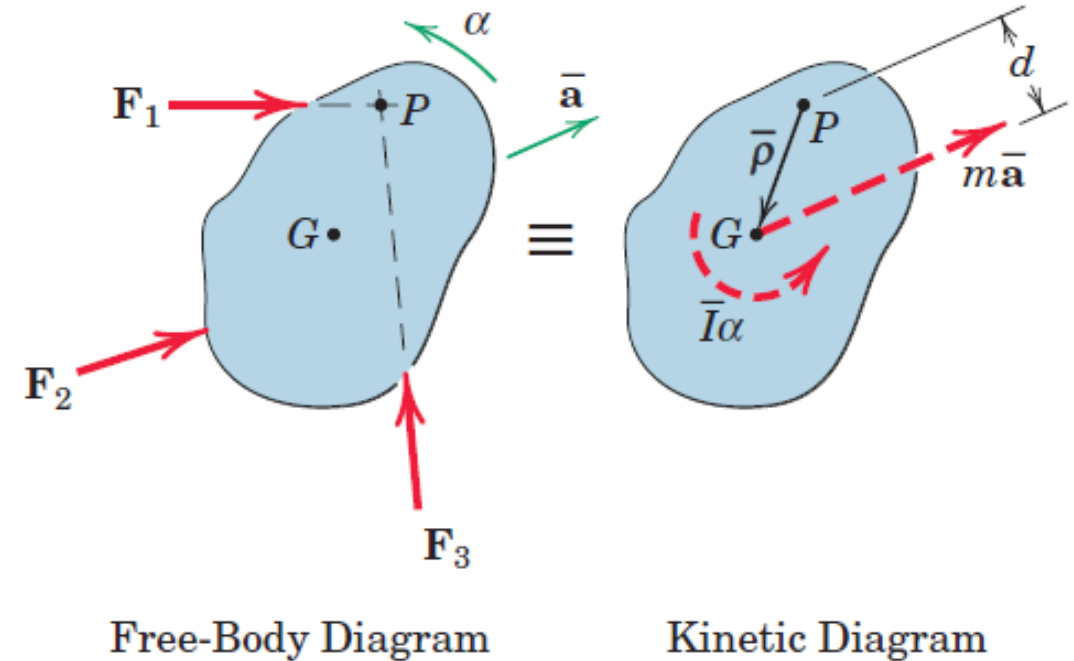


Alternate derivations

- If we had wished to eliminate reference to F_2 and F_3 , for example, by choosing their intersection as the reference point, then P would lie on the opposite side of the $m\bar{a}$ vector, and the clockwise moment of $m\bar{a}$ about P would be a negative term in the equation

$$\sum M_P = \bar{I}\alpha + m\bar{a}d$$

- It is merely an expression of the familiar principle of moments, where the sum of the moments about P equals the combined moment about P of their sum, expressed by the resultant couple $\sum M_G = \bar{I}\alpha$ and the resultant force $\sum F = m\bar{a}$



Alternate derivations

➤ Derived Earlier

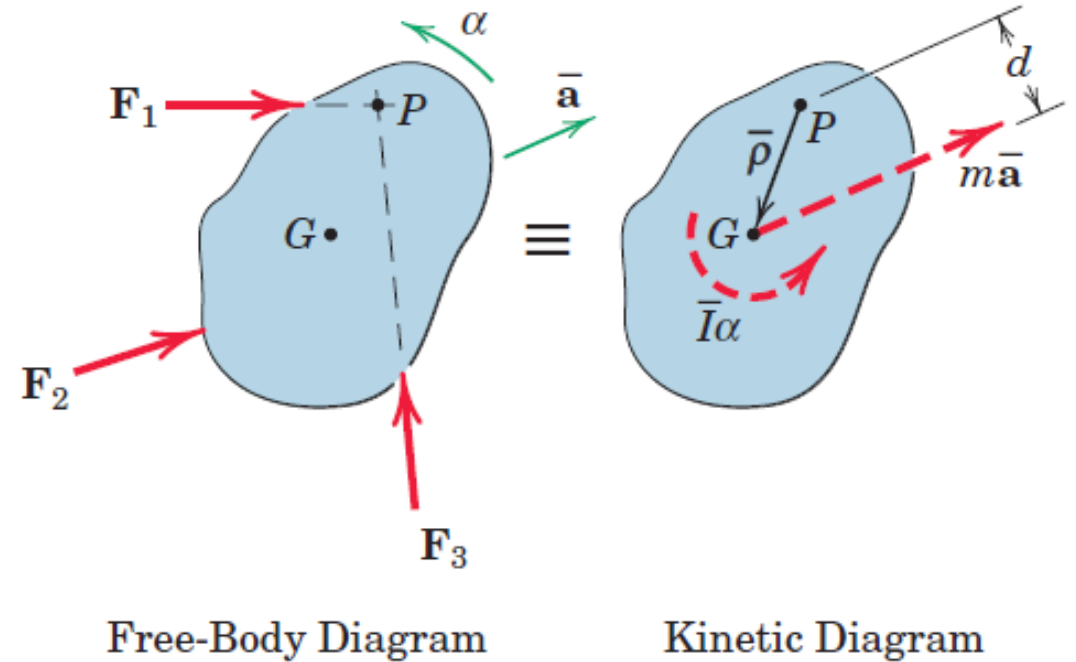
$$\sum \mathbf{M}_p = (\dot{\mathbf{H}}_p)_{rel} + \bar{\rho} \times m \mathbf{a}_p$$

➤ If point P is chosen as fixed to the body,

$$(\dot{\mathbf{H}}_p)_{rel} = I_P \alpha$$

Where I_P is the mass moment of inertia about an axis through P and α is the angular acceleration of the body

$$\sum \mathbf{M}_p = I_P \alpha + \bar{\rho} \times m \mathbf{a}_p$$







Alternate derivations

➤ If $\bar{\rho} = 0$

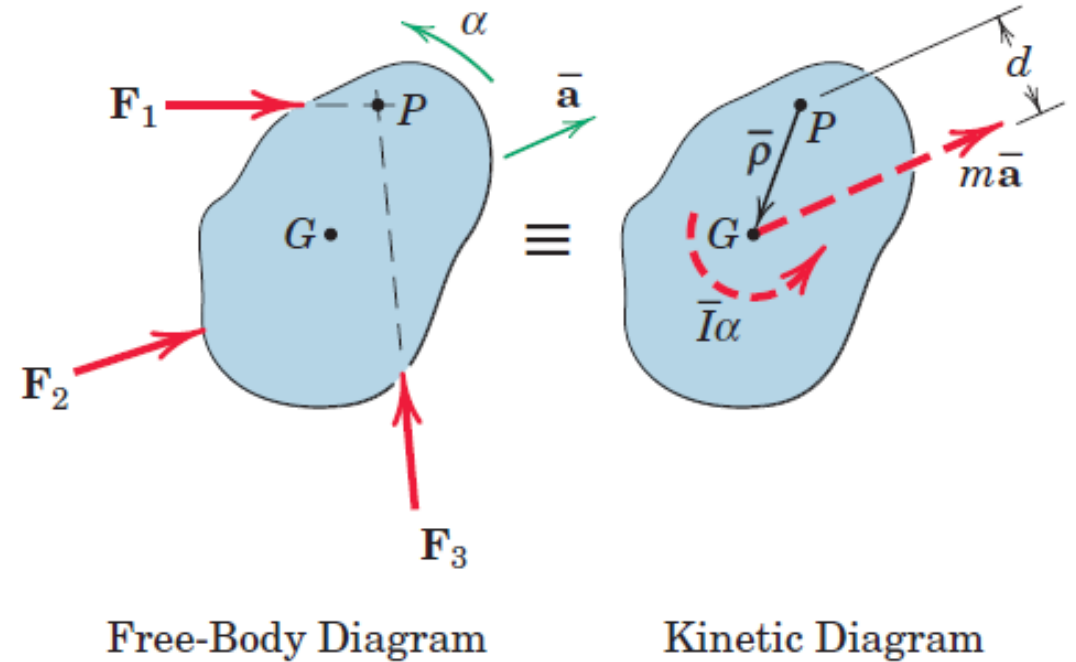
$$\sum \mathbf{M}_P = I_P \alpha + \bar{\rho} \times m \mathbf{a}_P$$

$$\sum \mathbf{M}_G = I_G \alpha$$

➤ If $\mathbf{a}_P = 0$,

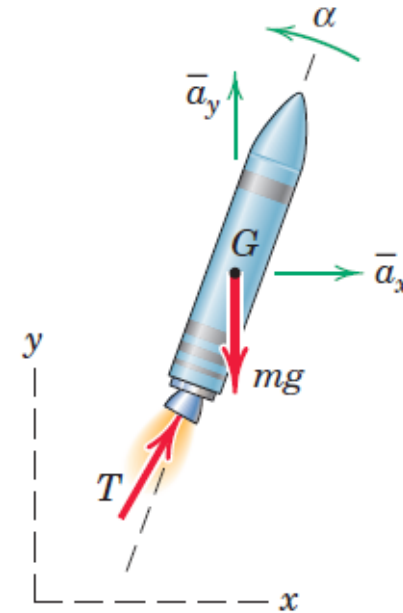
$$\sum \mathbf{M}_P = I_P \alpha$$

➤ $\mathbf{a}_P = 0$ will happen if body is rotating about a non-accelerating point P .

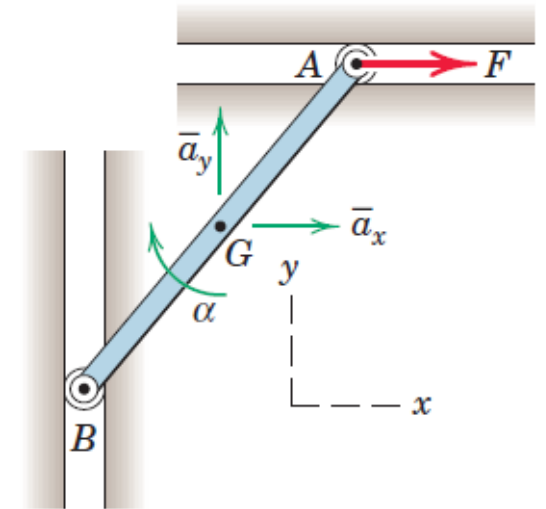


Unconstrained and Constrained Motion

- The rocket moving in a vertical plane is an example of unconstrained motion as there are no physical confinements to its motion
- The two components a_x and a_y of the mass-center acceleration and the angular acceleration α may be determined independently of one another.
- The bar undergoes a constrained motion, where the vertical and horizontal guides for the ends of the bar impose a kinematic relationship between the acceleration components of the mass center and the angular acceleration of the bar.
- Thus, it is necessary to determine this kinematic relationship from the principles established in Chapter 5 and to combine it with the force and moment equations of motion before a solution can be carried out.



(a) Unconstrained Motion

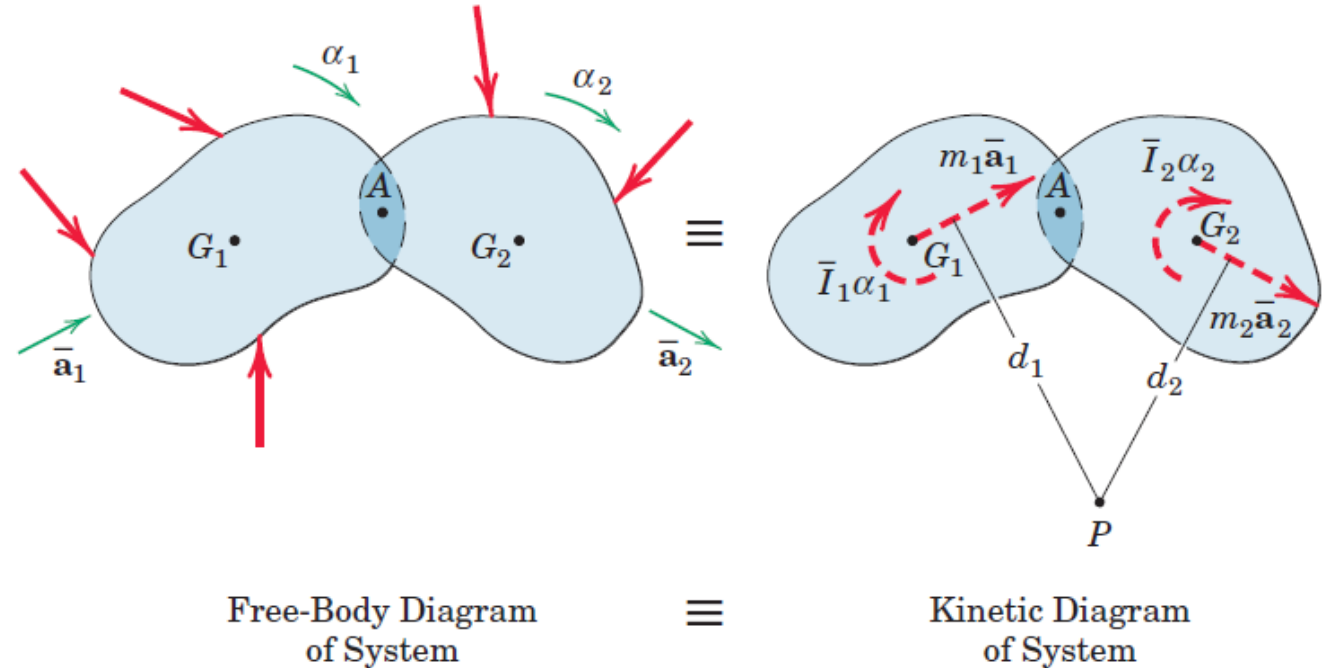


(b) Constrained Motion

System of interconnected Bodies

- Figure illustrates two rigid bodies hinged at A and subjected to the external forces
- The forces in the connection at A are internal to the system and are not disclosed. The resultant of all external forces must equal the vector sum of the two resultants $m_1 \bar{a}_1$ and $m_2 \bar{a}_2$, and the sum of the moments about some arbitrary point such as P of all external forces must equal the moment of the resultants, $\bar{I}_1 \alpha_1 + \bar{I}_2 \alpha_2 + m_1 \bar{a}_1 d_1 + m_2 \bar{a}_2 d_2$

$$\sum \mathbf{F} = \sum m \bar{\mathbf{a}} \quad \sum \mathbf{M}_G = \sum \bar{I} \alpha$$

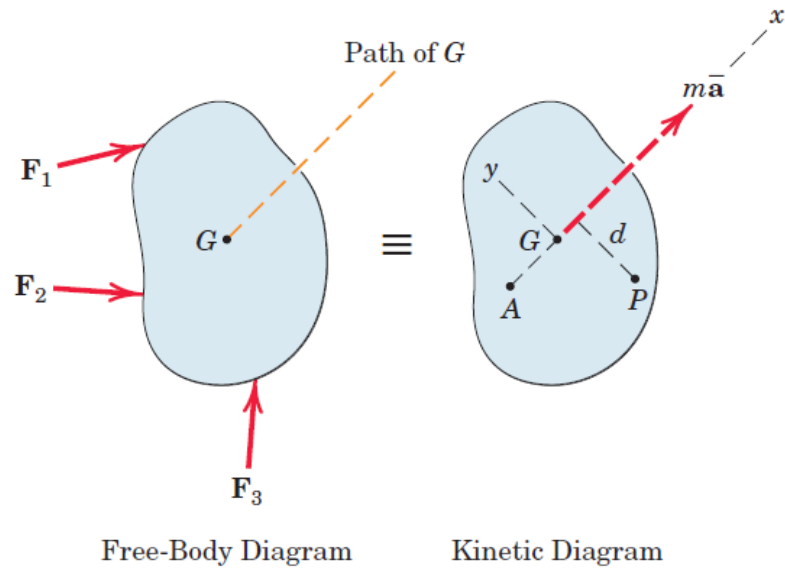


Summations on the right-hand side of the equations represent as many terms as there are separate bodies

Translation

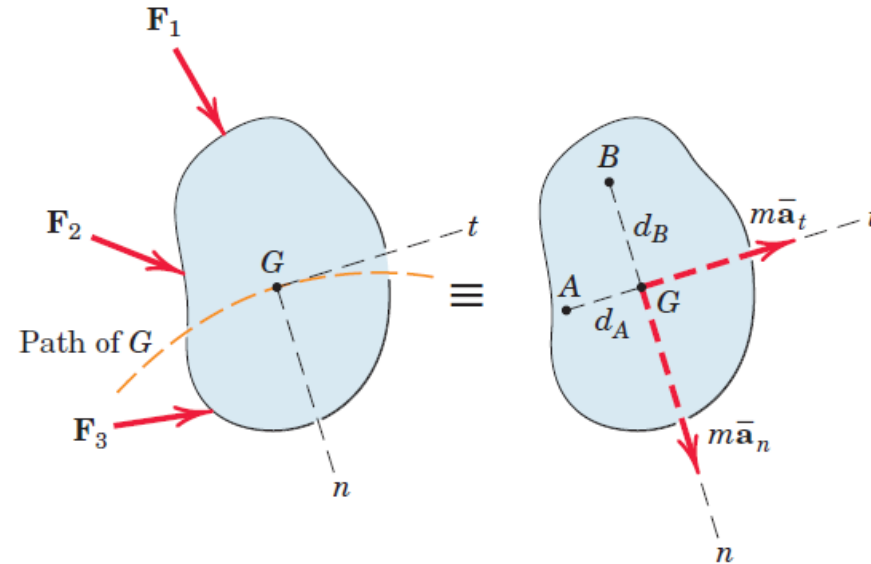
- Every line in a translating body remains parallel to its original position at all times. In rectilinear translation all points move in straight lines, whereas in curvilinear translation all points move on congruent curved paths.
- In either case, there is no angular motion of the translating body, so that both ω and α are zero.
- All reference to the moment of inertia is eliminated for a translating body

Translation



(a) Rectilinear Translation
($\alpha = 0, \omega = 0$)

$$\sum F = m\bar{a}$$

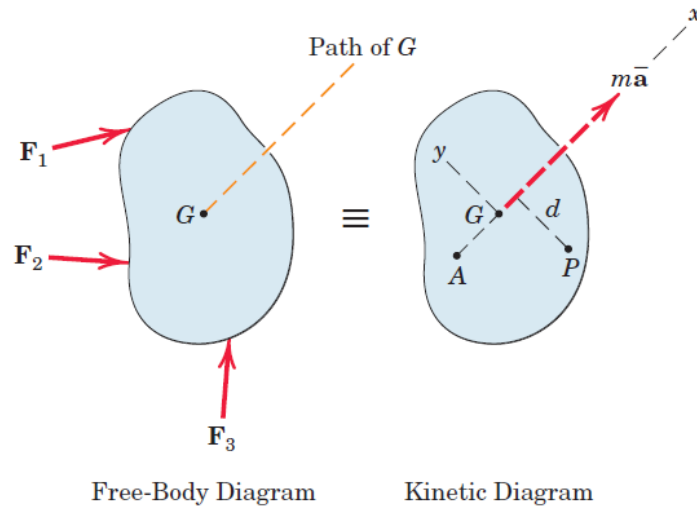


Free-Body Diagram Kinetic Diagram

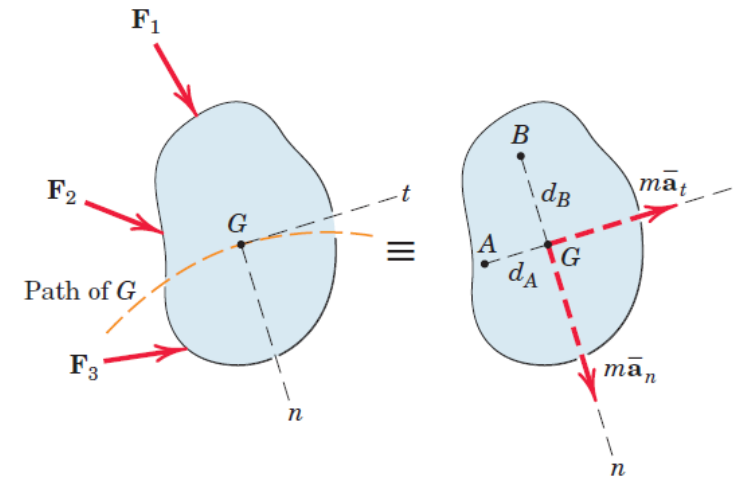
(b) Curvilinear Translation
($\alpha = 0, \omega = 0$)

$$\sum M_G = \bar{I}\alpha = 0$$

Translation



(a) Rectilinear Translation
($\alpha = 0, \omega = 0$)



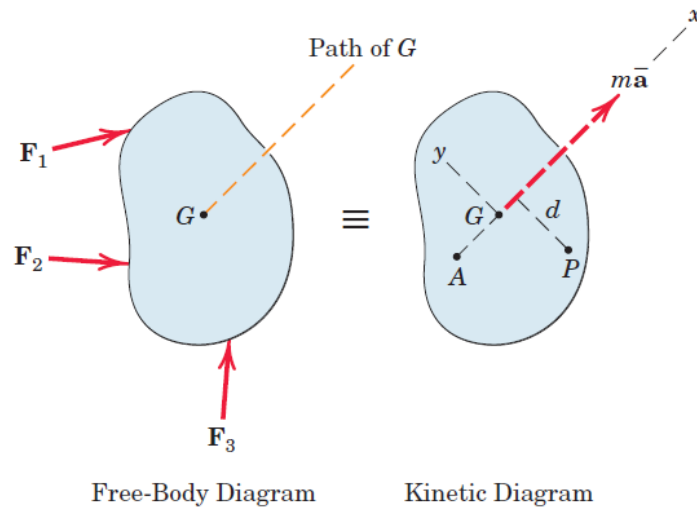
(b) Curvilinear Translation
($\alpha = 0, \omega = 0$)

$$\sum F_x = m\bar{a}_x \quad \sum F_y = m\bar{a}_y = 0$$

$$\sum F_n = m\bar{a}_n \quad \sum F_t = m\bar{a}_t$$

$$\sum M_G = \bar{I}\alpha = 0$$

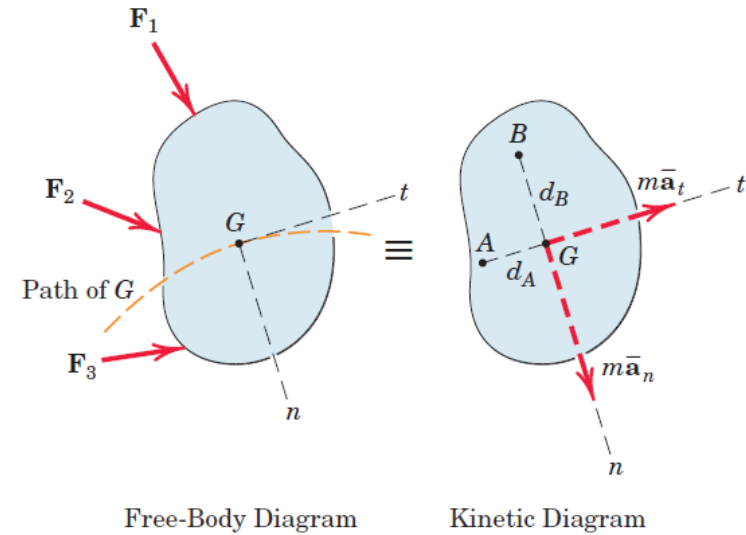
Translation



(a) Rectilinear Translation
($\alpha = 0, \omega = 0$)

$$\sum M_P = \bar{I}\alpha + m\bar{a}d$$

$$\sum M_P = m\bar{a}d \quad \sum M_A = 0$$

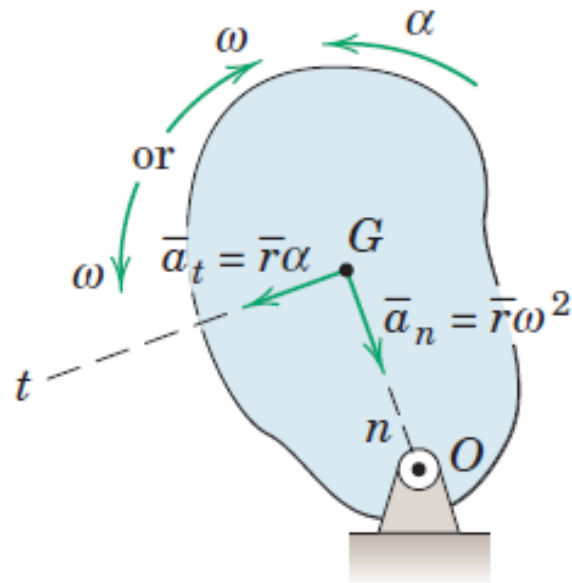


(b) Curvilinear Translation
($\alpha = 0, \omega = 0$)

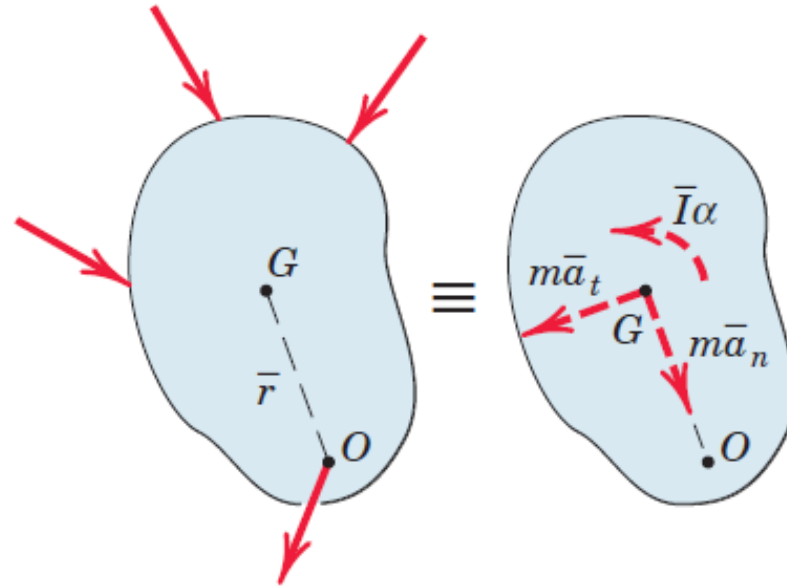
$$\sum M_A = m\bar{a}_n d_A \quad \sum M_B = m\bar{a}_t d_B$$

Fixed Axis Rotation

- All points in the body describe circles about the rotation axis, and all lines of the body in the plane of motion have the same angular velocity ω and angular acceleration α



Fixed-Axis Rotation



Free-Body Diagram

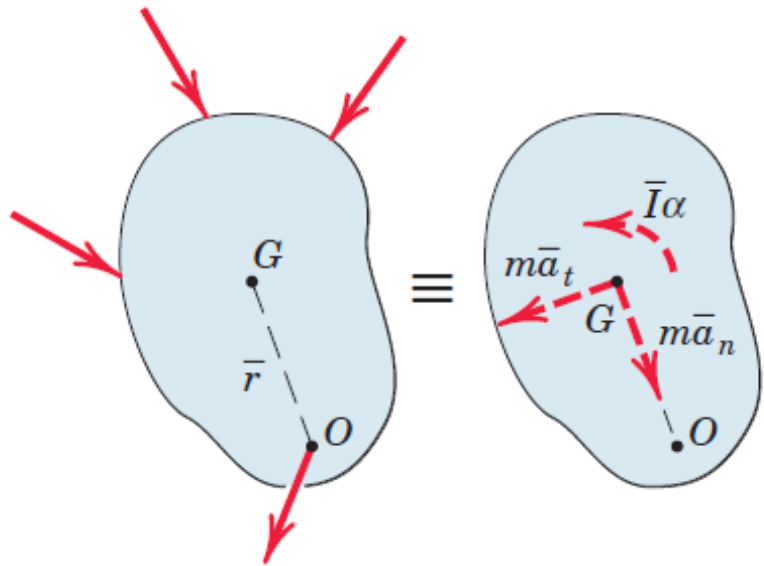
Kinetic Diagram

Fixed Axis Rotation

$$\sum F_n = m\bar{a}_n = m\bar{r}\omega^2$$

$$\sum F_t = m\bar{a}_t = m\bar{r}\alpha$$

$$\sum M_G = \bar{I}\alpha$$



Free-Body Diagram

Kinetic Diagram

- In applying the moment equation about G , one must account for the moment of the force applied to the body at O

Fixed Axis Rotation

$$\sum M_o = \bar{I}\alpha + m\bar{a}_t\bar{r}$$

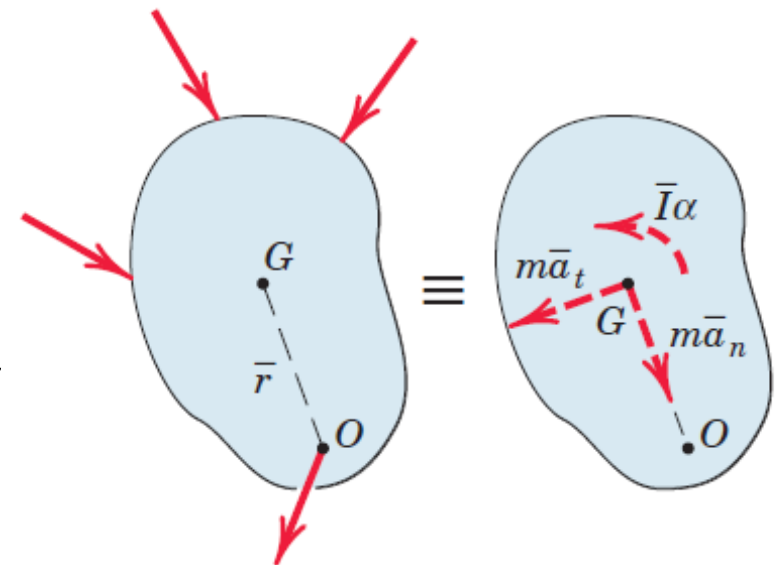
➤ Moment equation about the rotation axis O

$$I_o = \bar{I} + m\bar{r}^2 \quad I_o - m\bar{r}^2 = \bar{I}$$

$$\sum M_o = (I_o - m\bar{r}^2)\alpha + m\bar{a}_t\bar{r}$$

$$\sum M_o = I_o\alpha - m\bar{r}^2\alpha + m\bar{a}_t\bar{r} = I_o\alpha$$

$$\sum M_o = I_o\alpha$$



Free-Body Diagram

Kinetic Diagram





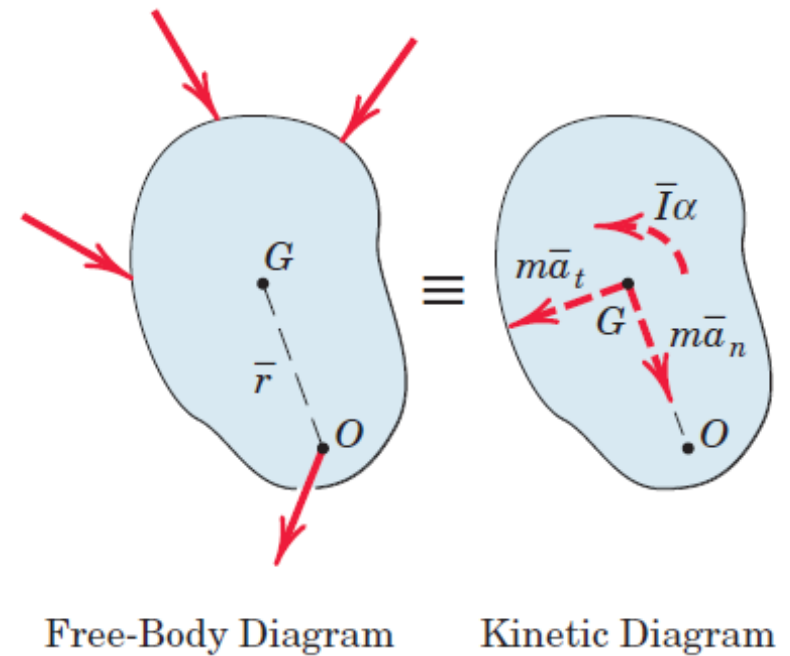
Fixed Axis Rotation

- Moment equation about the rotation axis O

For the case of rotation of a rigid body about a fixed axis through its mass center G ,

$$\bar{a} = 0 \quad \sum F = 0$$

The resultant of the applied forces then is the couple $\bar{I}\alpha$



Fixed Axis Rotation

$$\sum M_o = \bar{I}\alpha + m\bar{a}_t\bar{r}$$

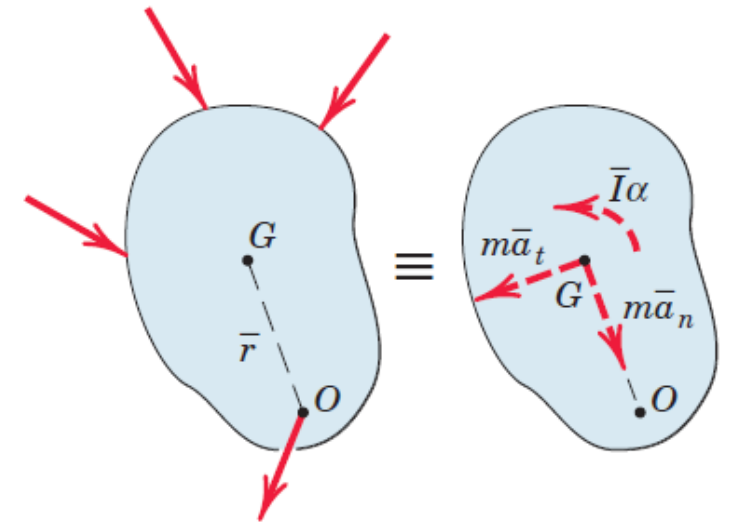
➤ Shift $m\bar{r}\alpha$ to point Q

$$\text{➤ } m\bar{r}\alpha q = \bar{I}\alpha + m\bar{r}\alpha\bar{r} = (I_o - m\bar{r}^2)\alpha + m\bar{r}\alpha\bar{r} = k_o^2 m\alpha$$

$$\text{➤ } m\bar{r}\alpha q = k_o^2 m\alpha$$

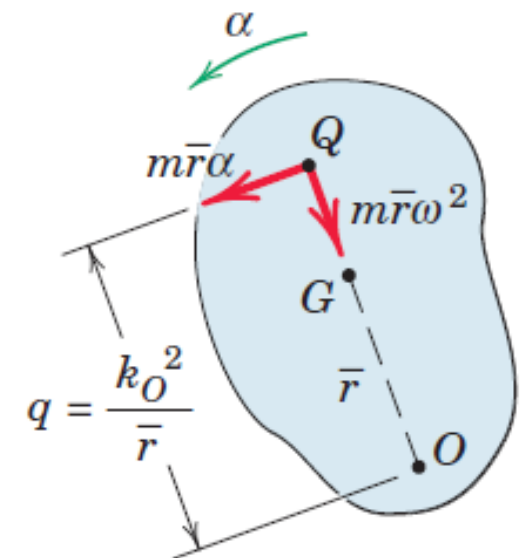
$$\text{➤ } q = \frac{k_o^2}{\bar{r}}$$

- Point Q is called the **center of percussion** and has the unique property that the **resultant of all forces applied to the body must pass through it**.
- It follows that the sum of the moments of all forces about the center of percussion is always zero, $\sum M_Q = 0$



Free-Body Diagram

Kinetic Diagram





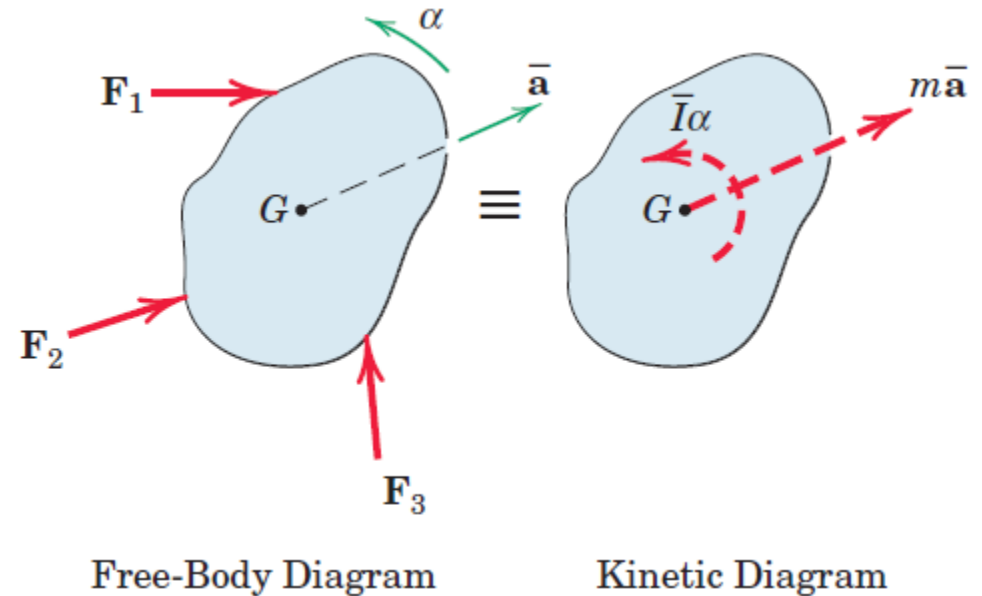


General Plane Motion

$$\sum F = m\bar{a}$$

$$\sum M_G = \bar{I}\alpha$$

- Direct application of these equations expresses the equivalence between the externally applied forces, as disclosed by the free-body diagram, and their force and moment resultants, as represented by the kinetic diagram



Work-Energy Relation: Work of Forces and Couples

Work done by a force $\mathbf{F}$

$$U = \int \mathbf{F} \cdot d\mathbf{r}$$

$$U = \int \mathbf{F} \cos \alpha \, ds$$

$d\mathbf{r}$ is the infinitesimal vector displacement of the point of application

α is the angle between $\mathbf{F}$ and the direction of the displacement, $d\mathbf{r}$

ds is the magnitude of the vector displacement $d\mathbf{r}$

Work-Energy Relation: Work of Couples

Work done by a couple M which acts on a rigid body during its motion

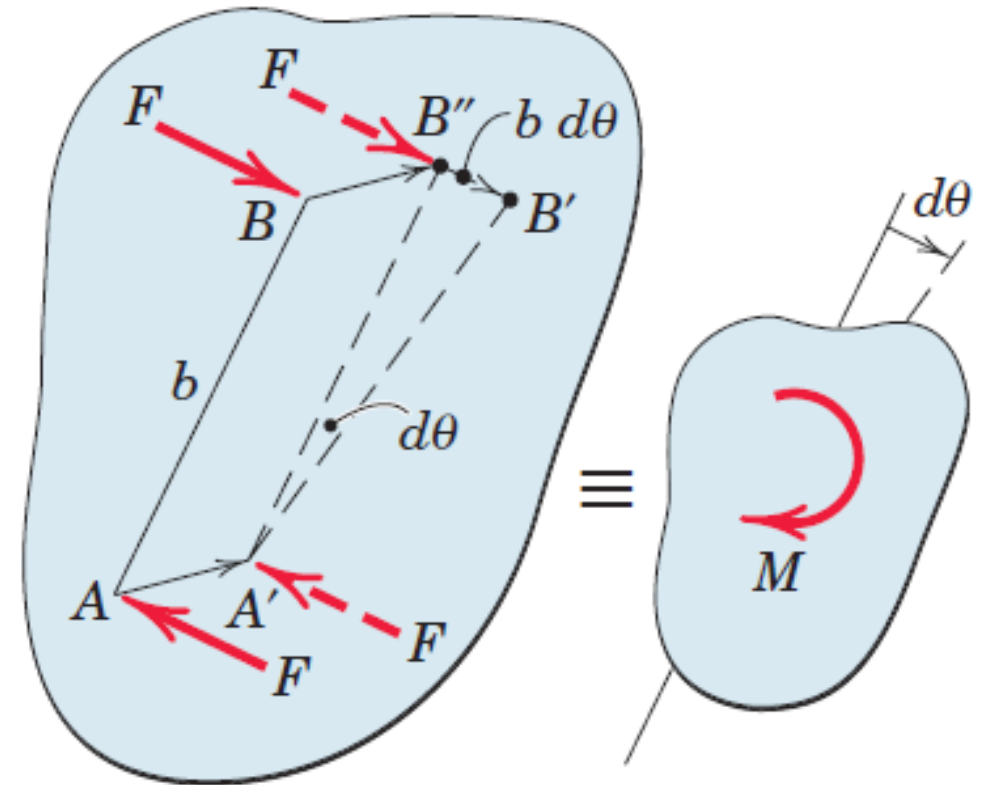
$$U = \int M d\theta$$

Body moves from AB to $A'B''$ to $A'B'$.

During translation from AB to $A'B''$, Work done by one force cancels work done by another force

Work is done during rotational motion only

$$dU = Fb d\theta = Md\theta$$

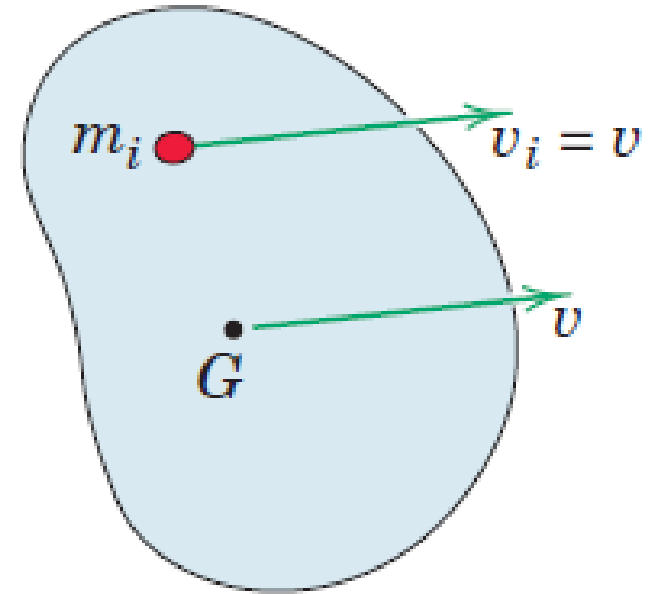


Kinetic Energy: Translation

Kinetic Energy of a particle of mass m_i

$$T_i = \frac{1}{2} m_i v_i^2$$

$$T = \sum T_i = \sum \frac{1}{2} m_i v_i^2 = \sum \frac{1}{2} m_i v^2 = \frac{1}{2} v^2 \sum m_i = \frac{1}{2} m v^2$$



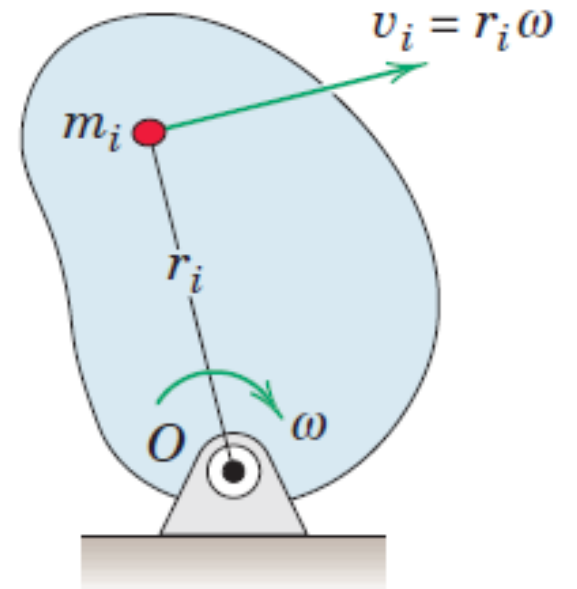
(a) Translation

Kinetic Energy: Fixed Axis Rotation

Kinetic Energy of a particle of mass m_i

$$T_i = \frac{1}{2} m_i \omega^2 r_i^2$$

$$\begin{aligned} T &= \sum T_i = \sum \frac{1}{2} m_i v_i^2 = \sum \frac{1}{2} m_i \omega^2 r_i^2 \\ &= \frac{1}{2} \omega^2 \sum m_i r_i^2 = \frac{1}{2} I_o \omega^2 \end{aligned}$$



(b) Fixed-Axis Rotation



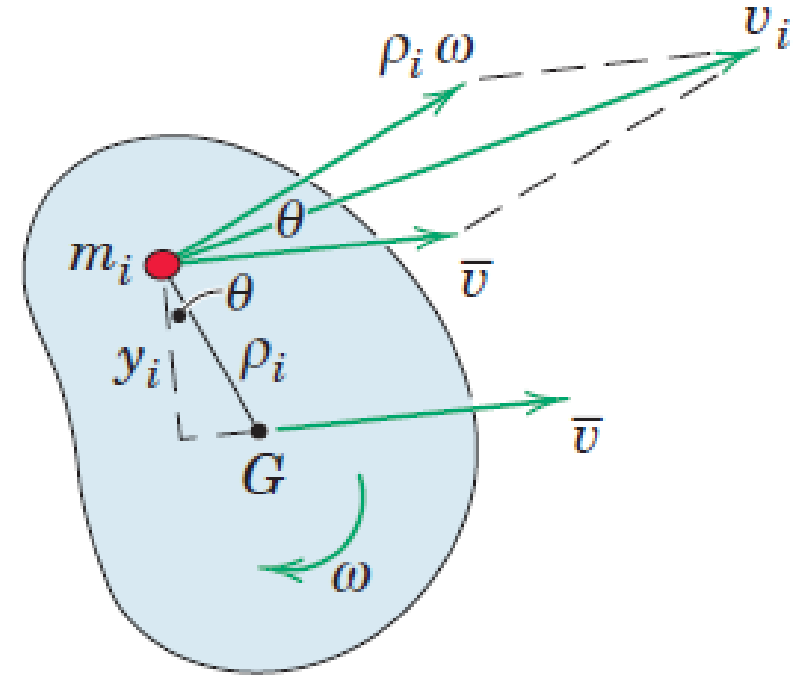


Kinetic Energy: General Plane Motion

Kinetic Energy of a particle of mass m_i

$$T_i = \frac{1}{2} m_i v_i^2$$

$$\begin{aligned} T &= \sum T_i = \sum \frac{1}{2} m_i v_i^2 \\ &= \sum \frac{1}{2} m_i (\bar{v}^2 + (\rho_i \omega)^2 + 2\bar{v} \rho_i \omega \cos \theta) \end{aligned}$$



(c) General Plane Motion

Kinetic Energy: General Plane Motion

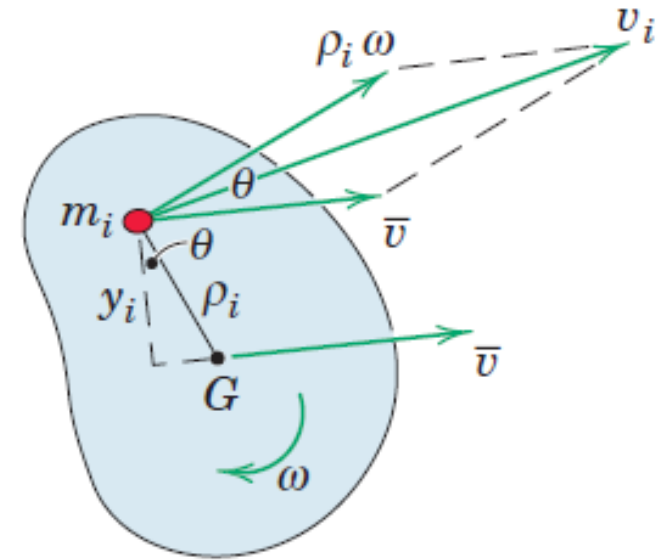
Kinetic Energy of a particle of mass m_i

$$T = \sum T_i = \sum \frac{1}{2} m_i v_i^2 = \sum \frac{1}{2} m_i (\bar{v}^2 + (\rho_i \omega)^2 + 2\bar{v} \rho_i \omega \cos \theta)$$

$$T = \sum \frac{1}{2} m_i (\bar{v}^2 + (\rho_i \omega)^2 + 2\bar{v} \rho_i \omega \cos \theta)$$

$$T = \sum \frac{1}{2} m_i (\bar{v}^2 + (\rho_i \omega)^2) + \sum m_i \bar{v} \rho_i \omega \cos \theta$$

$$\sum m_i \bar{v} \rho_i \omega \cos \theta = \sum m_i \bar{v} \omega \rho_i \cos \theta = \bar{v} \omega \sum y_i m_i = 0$$



(c) General Plane Motion





Kinetic Energy: General Plane Motion

Kinetic Energy of a particle of mass m_i

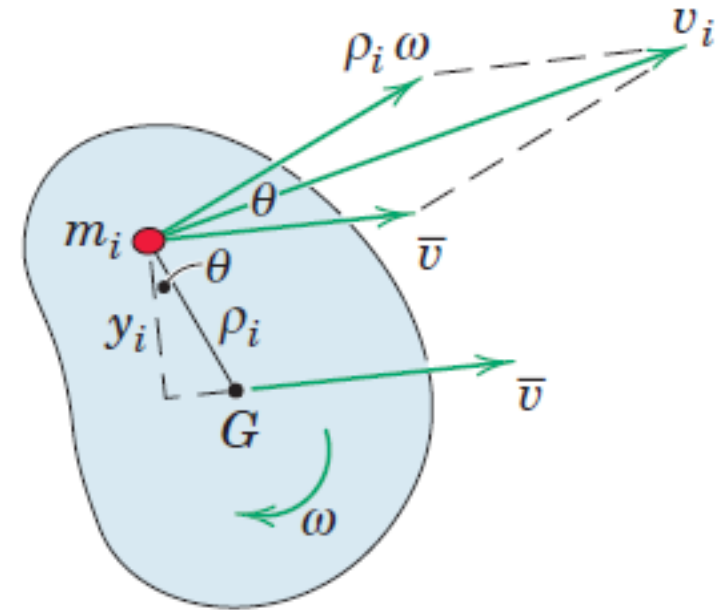
$$T = \sum \frac{1}{2} m_i (\bar{v}^2 + (\rho_i \omega)^2)$$

$$T = \sum \frac{1}{2} m_i \bar{v}^2 + \sum \frac{1}{2} m_i (\rho_i \omega)^2$$

$$T = \frac{1}{2} \bar{v}^2 \sum m_i + \frac{1}{2} (\omega)^2 \sum m_i (\rho_i)^2$$

$$T = \frac{1}{2} m \bar{v}^2 + \frac{1}{2} \bar{I} (\omega)^2$$

$\bar{I}$ is mass moment of Inertia about its mass center



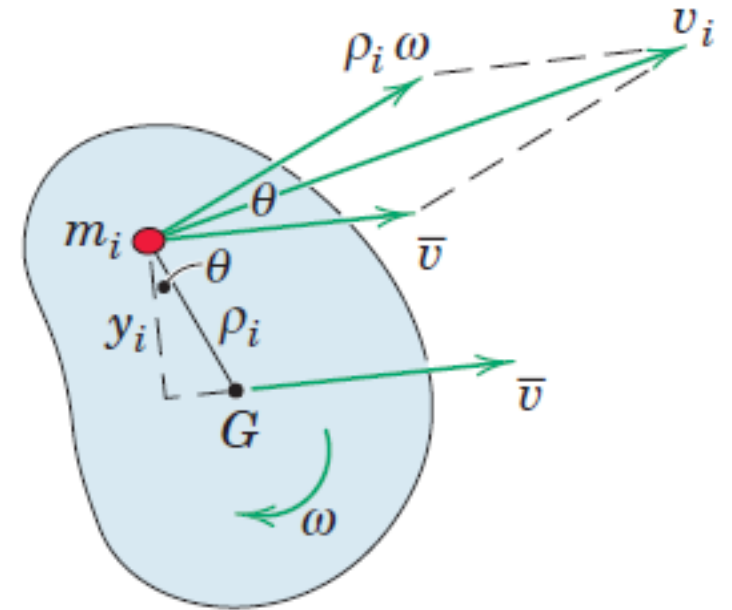
(c) General Plane Motion

Kinetic Energy: General Plane Motion

Kinetic Energy about instantaneous center of zero velocity

$$T = \frac{1}{2} I_C (\omega)^2$$

I_C is mass moment of Inertia about instantaneous center of zero velocity



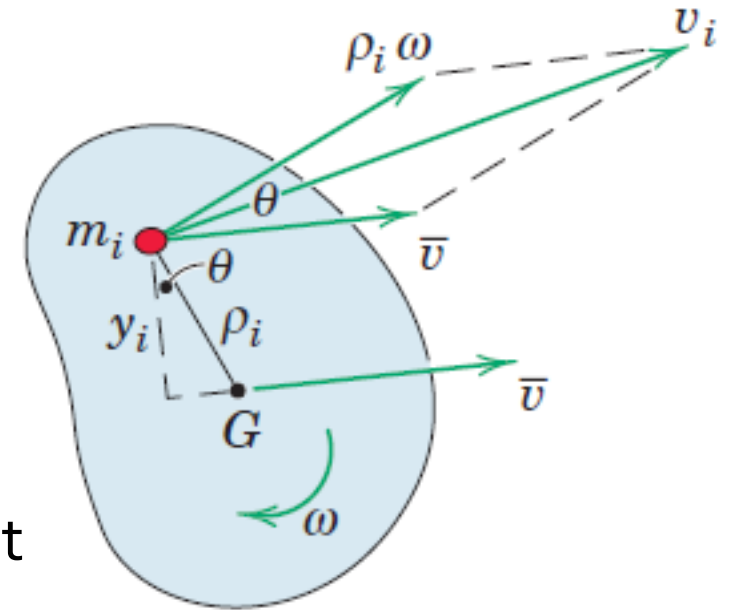
(c) General Plane Motion

Potential Energy and Work Energy Equation

$$T_2 = U_{1-2} + T_1$$

$$T_2 + V_2 = U'_{1-2} + T_1 + V_1$$

prime denotes the work done by all forces other than weight and spring forces



(c) General Plane Motion

Power

Power is the time rate at which work is performed

$$P = \frac{dU}{dt} = \frac{\mathbf{F} \cdot d\mathbf{r}}{dt} = \mathbf{F} \cdot \mathbf{v}$$

$d\mathbf{r}$: differential displacement $\mathbf{v}$: velocity of point of application

$$P = \frac{dU}{dt} = \frac{\mathbf{M} \cdot d\theta}{dt} = \mathbf{M} \cdot \boldsymbol{\omega}$$

$d\theta$: differential angular displacement $\boldsymbol{\omega}$: angular velocity

- If the senses of M and $\boldsymbol{\omega}$ are the same, the power is positive and energy is supplied to the body.
- Conversely, if M and $\boldsymbol{\omega}$ have opposite senses, the power is negative and energy is removed from the body.

Power

Power is the time rate at which work is performed

$$P = \frac{dU}{dt} = \frac{\mathbf{F} \cdot d\mathbf{r}}{dt} = \mathbf{F} \cdot \mathbf{v}$$

$d\mathbf{r}$: differential displacement $\mathbf{v}$: velocity of point of application

$$P = \frac{dU}{dt} = \frac{\mathbf{M} \cdot d\theta}{dt} = \mathbf{M} \cdot \boldsymbol{\omega}$$

$d\theta$: differential angular displacement $\boldsymbol{\omega}$: angular velocity

If the force $\mathbf{F}$ and the couple $\mathbf{M}$ act simultaneously, the total instantaneous power is

$$P = \mathbf{F} \cdot \mathbf{v} + \mathbf{M} \cdot \boldsymbol{\omega}$$

Power

Power by evaluating the rate at which the total mechanical energy of a rigid body or a system of rigid bodies is changing

$$dU' = dT + dV$$

dU' work of the active forces and couples applied to the body or to the system of bodies

dV accounts for work by gravitational forces and that of spring forces

$$\frac{dU'}{dt} = \frac{dT}{dt} + \frac{dV}{dt} = \dot{T} + \dot{V}$$

Power developed by the active forces and couples equals the rate of change of the total mechanical energy of the body or system of bodies

Power

$$\dot{T} = \frac{dT}{dt} = \frac{d}{dt} \left(\frac{1}{2} m \bar{\mathbf{v}} \cdot \bar{\mathbf{v}} + \frac{1}{2} \bar{I} \omega^2 \right)$$

$$\dot{T} = \frac{dT}{dt} = \frac{d}{dt} \left(\frac{1}{2} m \bar{\mathbf{v}} \cdot \bar{\mathbf{v}} + \frac{1}{2} \bar{I} \omega^2 \right)$$

$$\dot{T} = \frac{dT}{dt} = (m \bar{\mathbf{a}} \cdot \bar{\mathbf{v}} + \bar{I} \omega \dot{\omega}) = (m \bar{\mathbf{a}} \cdot \bar{\mathbf{v}} + \bar{I} \omega \alpha) = (\mathbf{R} \cdot \bar{\mathbf{v}} + \bar{M} \omega)$$

$\mathbf{R}$ is the **resultant of all forces acting on the body** and $\bar{M}$ is the **resultant moment about the mass center G of all forces**.

Acceleration from Work Energy: Virtual Work

Work-Energy Equation

$$dU' = dT + dV$$

- The term dU' represents the total work done by all active nonpotential forces acting on the system under consideration during the infinitesimal displacement of the system
- The work of potential forces is included in the dV –term. If we use the subscript i to denote a representative body of the interconnected system, the differential change in kinetic energy T for the entire system becomes

$$dT = d\left(\sum \frac{1}{2} m_i \bar{v}_i^2 + \sum \frac{1}{2} \bar{I}_i \omega_i^2\right) = \left(\sum m_i \bar{v}_i d\bar{v}_i + \sum \bar{I}_i \omega_i d\omega_i\right)$$

$d\bar{v}_i$ and $d\omega_i$ are respective changes in velocities

Acceleration from Work Energy: Virtual Work

Work-Energy Equation

$$dU' = dT + dV$$
$$dT = d \left(\sum \frac{1}{2} m_i \bar{v}_i^2 + \sum \frac{1}{2} \bar{I}_i \omega_i^2 \right) = \left(\sum m_i \bar{v}_i d\bar{v}_i + \sum \bar{I}_i \omega_i d\omega_i \right)$$

$d\bar{v}_i$ and $d\omega_i$ are respective changes in velocities

$$m_i \bar{v}_i d\bar{v}_i = m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i$$
$$\bar{I}_i \omega_i d\omega_i = \bar{I}_i \alpha_i d\theta_i$$

$d\bar{\mathbf{s}}_i$ and $d\theta_i$ are infinitesimal linear displacement of center of mass and infinitesimal angular displacement of the body in the plane of motion

$$dT = \left(\sum m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i + \sum \bar{I}_i \alpha_i d\theta_i \right)$$

Acceleration from Work Energy: Virtual Work

Work-Energy Equation

$$dU' = dT + dV$$

$$dT = \left(\sum m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i + \sum \bar{I}_i \alpha_i d\theta_i \right)$$
$$dT = \left(\sum \mathbf{R}_i \cdot d\bar{\mathbf{s}}_i + \sum \mathbf{M}_{G_i} \cdot d\boldsymbol{\theta}_i \right)$$

$\mathbf{R}_i$ and $\mathbf{M}_{G_i}$ are resultant force and resultant couple acting on body

$$d\boldsymbol{\theta}_i = d\theta_i \mathbf{k}$$

- Differential change in kinetic energy equals the differential work done on the system by the resultant forces and resultant couples acting on all the bodies of the system

Acceleration from Work Energy: Virtual Work

The term dV represents the differential change in the total gravitational potential energy V_g and the total elastic potential energy V_e

$$dV = d \left(\sum m_i g h_i + \sum \frac{1}{2} k_j x_j^2 \right) = \sum m_i g dh_i + \sum k_j x_j dx_j$$

- h_i represents the vertical distance of the center of mass of the representative body of mass m_i above any convenient datum plane
- x_j stands for the deformation, tensile or compressive, of a representative elastic member of the system (spring) whose stiffness is k_j

$$dU' = \sum m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i + \sum \bar{I}_i \alpha_i d\theta_i + \sum m_i g dh_i + \sum k_j x_j dx_j$$

Acceleration from Work Energy: Virtual Work

$$dU' = \sum m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i + \sum \bar{I}_i \alpha_i d\theta_i + \sum m_i g dh_i + \sum k_j x_j dx_j$$

- $m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i$ and $\bar{I}_i \alpha_i d\theta_i$ will be positive if the accelerations are in the same direction as the respective displacements and negative if they are in the opposite direction
- Equation has the advantage of relating the accelerations to the active forces directly, which eliminates the need for dismembering the system and then eliminating the internal forces and reactive forces by simultaneous solution of the force-mass-acceleration equations for each member

Virtual Work

$$dU' = \sum m_i \bar{\mathbf{a}}_i \cdot d\bar{\mathbf{s}}_i + \sum \bar{I}_i \alpha_i d\theta_i + \sum m_i g dh_i + \sum k_j x_j dx_j$$

In above equation, the differential motions are differential changes in the real or actual displacements which occur

- For a mechanical system which assumes a steady-state configuration during constant acceleration, we often find it convenient to introduce the concept of virtual work.
- The concepts of virtual work and virtual displacement were introduced and used to establish equilibrium configurations for static systems of interconnected bodies

Virtual Work

$$\delta U' = \sum m_i \bar{\mathbf{a}}_i \cdot \delta \bar{\mathbf{s}}_i + \sum \bar{I}_i \alpha_i \delta \theta_i + \sum m_i g \delta h_i + \sum k_j x_j \delta x_j$$

- A virtual displacement is any assumed and arbitrary displacement, linear or angular, away from the natural or actual position.
- For a system of connected bodies, the virtual displacements must be consistent with the constraints of the system.
- For example, when one end of a link is hinged about a fixed pivot, the virtual displacement of the other end must be normal to the line joining the two ends.
- Such requirements for displacements consistent with the constraints are purely kinematic and provide what are known as the **equations of constraint**.

Virtual Work

$$\delta U' = \sum m_i \bar{\mathbf{a}}_i \cdot \delta \bar{\mathbf{s}}_i + \sum \bar{I}_i \alpha_i \delta \theta_i + \sum m_i g \delta h_i + \sum k_j x_j \delta x_j$$

- If a set of virtual displacements satisfying the equations of constraint and therefore consistent with the constraints is assumed for a mechanical system, the proper relationship between the coordinates which specify the configuration of the system will be determined by applying the work-energy relationship expressed in terms of virtual changes.
- It is customary to use the differential symbol d to refer to differential changes in the real displacements, whereas the symbol δ is used to signify virtual changes, that is, differential changes which are assumed rather than real.

Impulse-Momentum Equations

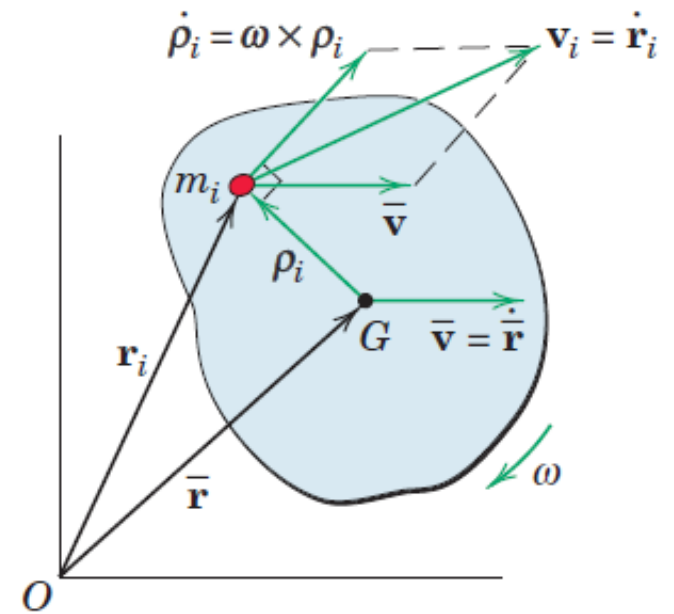
- Mass of particle m_i
- Momentum of particle $m_i v_i$
- Momentum of rigid body

$$G = \sum m_i v_i = \sum m_i \dot{r}_i = \frac{d \sum m_i r_i}{dt} = \frac{d (m \bar{r})}{dt} = m \bar{v}$$

where $\bar{v}$ is velocity of mass center

- Linear momentum of any mass system, rigid or non-rigid,

$$G = m \bar{v}$$



Impulse-Momentum Equations

- Mass of particle m_i
- Momentum of particle $m_i v_i$
- Momentum of rigid body

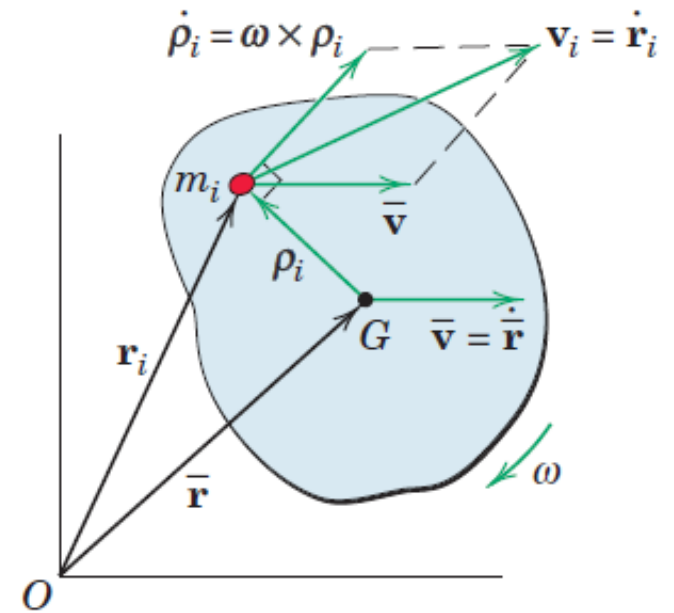
$$G = \sum m_i v_i = \sum m_i (\bar{\mathbf{v}} + \omega \times \rho_i) = \sum m_i \bar{\mathbf{v}} + \sum m_i \omega \times \rho_i$$

Where $\bar{\mathbf{v}}$ is velocity of mass center

$$G = \bar{\mathbf{v}} \sum m_i + \omega \times \sum m_i \rho_i = m \bar{\mathbf{v}} + 0$$

- Linear momentum of any mass system, rigid or non-rigid,

$$G = m \bar{\mathbf{v}}$$



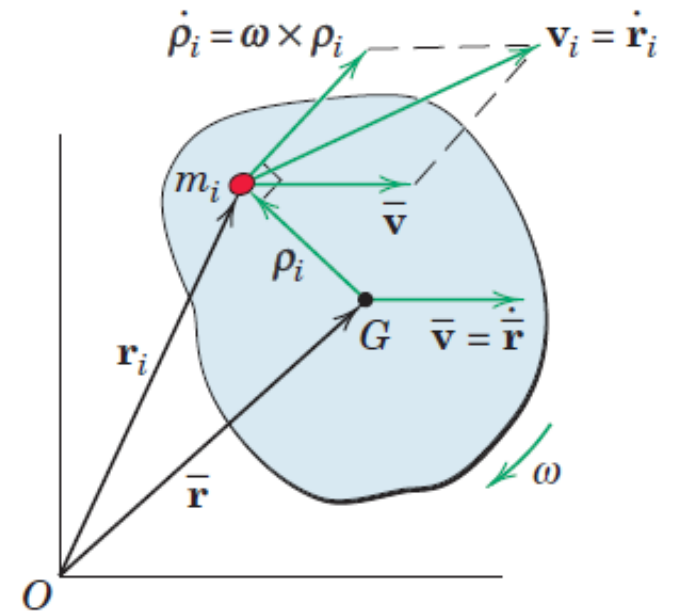
Impulse-Momentum Equations

$$\sum \mathbf{F} = \dot{\mathbf{G}}$$

Resultant force equals the time rate of change of momentum.

$$G_2 = \int_{t_1}^{t_2} F dt + G_1$$

Initial linear momentum plus the linear impulse acting on the body equals the final linear momentum.



Angular Momentum

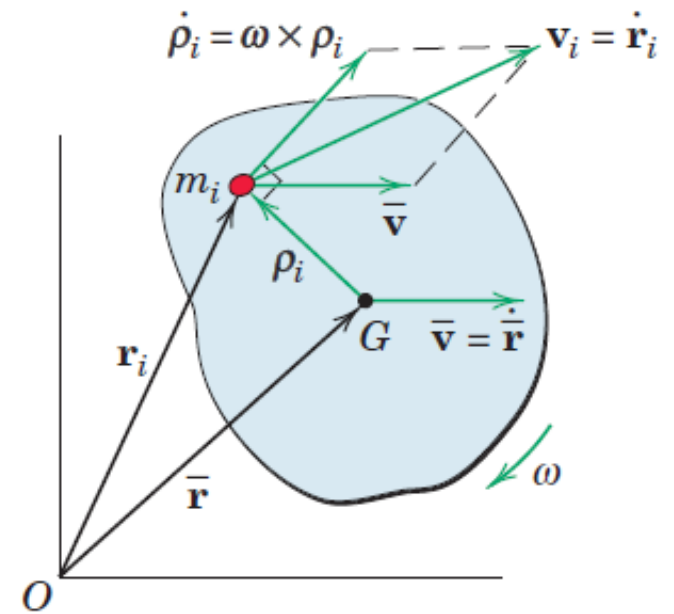
- Angular momentum is defined as the moment of linear momentum
- Angular momentum about the mass center of any prescribed system of mass

$$H_G = \sum \rho_i \times m_i v_i$$

Vector sum of the moments about G of the linear momenta of all particles.

$$H_G = \sum \rho_i \times m_i \dot{\rho}_i$$

$\dot{\rho}_i$ is velocity of mass m_i with respect to G.



Angular Momentum

Vector sum of the moments about G of the linear momenta of all particles.

$$H_G = \sum \rho_i \times m_i \dot{\rho}_i$$

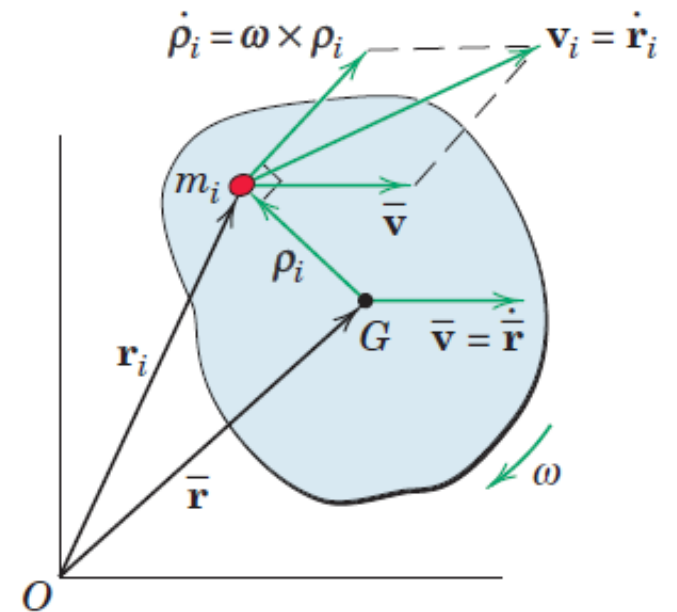
$\dot{\rho}_i$ is velocity of mass m_i with respect to G.

$$H_G = \sum \rho_i \times m_i (\omega \times \rho_i)$$

$$H_G = \sum \rho_i \times m_i (\omega \times \rho_i) = \sum m_i \rho_i \times \omega \times \rho_i = \sum \rho_i^2 m_i \omega \hat{\mathbf{k}} \\ = \bar{I} \omega$$

$$\bar{I} = \sum \rho_i^2 m_i$$

Mass moment of inertia of the body about its mass center



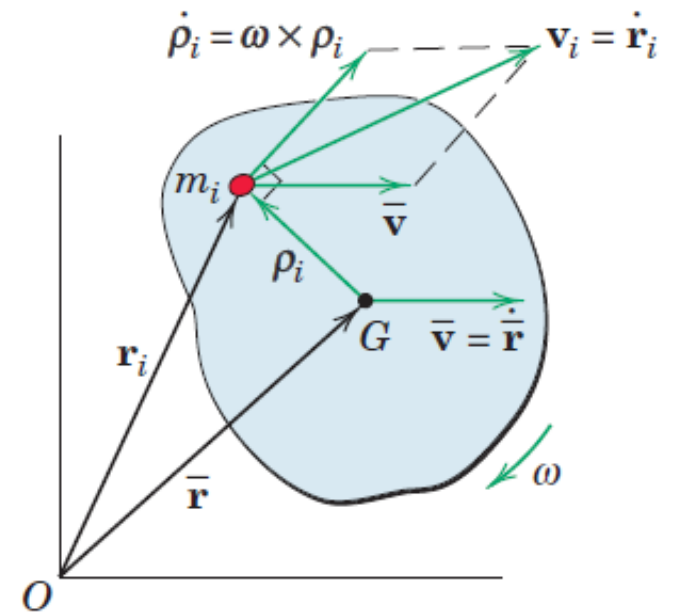
Angular Momentum

- Angular-momentum vector is always normal to the plane of motion
- Angular momentum about the mass center as the scalar

$$H_G = \bar{I}\omega$$

$$\sum M_G = \dot{H}_G$$

$$H_{G_2} = H_{G_1} + \int_{t_1}^{t_2} \sum M_G dt$$



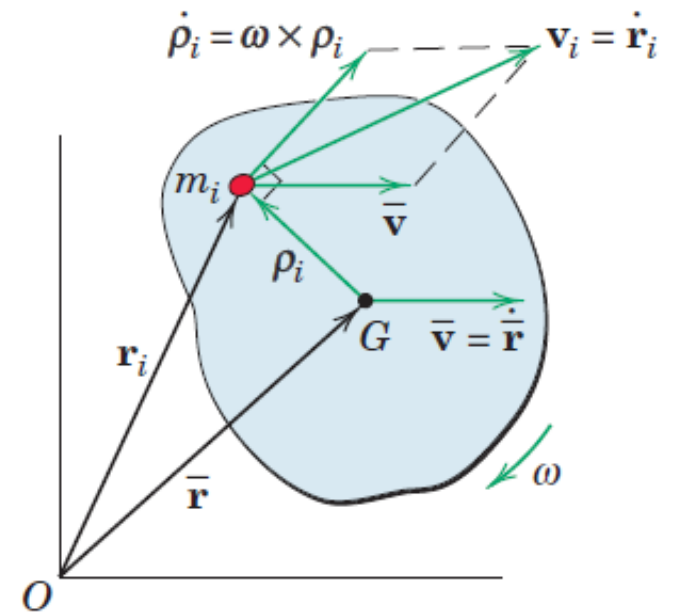
Angular Momentum

- Sum of the moments about the mass center of *all* forces acting on the body equals the time rate of change of angular momentum about the mass center.

$$\sum M_G = \dot{H}_G$$

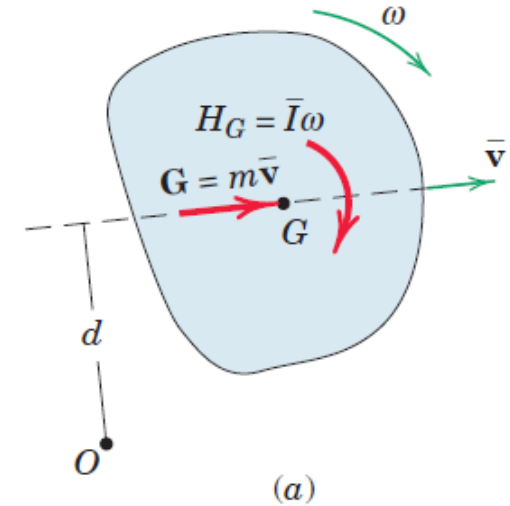
- Initial angular momentum about the mass center G plus the external angular impulse about G equals the final angular momentum about G

$$H_{G_2} = H_{G_1} + \int_{t_1}^{t_2} \sum M_G dt$$



Angular Momentum

- With the moments about G of the linear momenta of all particles accounted for by H_G it follows that we may represent the linear momentum $G = m\bar{v}$ as a vector through the mass center G
- G and H_G have vector properties analogous to those of the resultant force and couple.

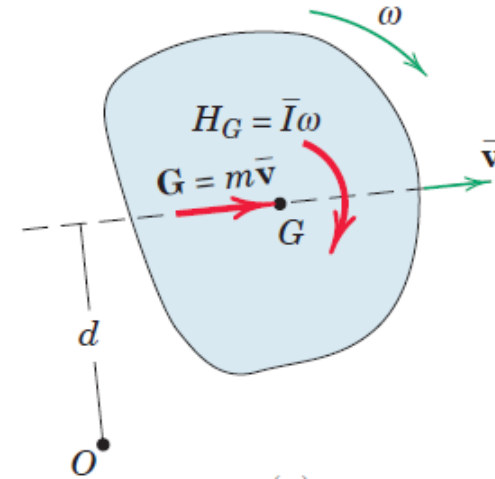


Angular Momentum

- Angular momentum H_O about any point O is

$$H_O = H_G + m \bar{v} d$$

- This expression holds at any particular instant of time about O, which may be a fixed or moving point on or off the body



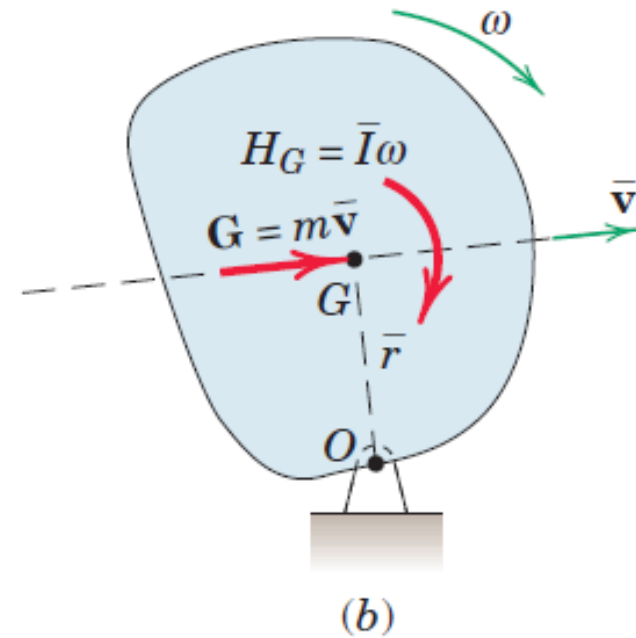
Angular Momentum

$$H_o = H_G + m \bar{\mathbf{v}} d$$

$$H_o = \bar{I}\omega + m \bar{\mathbf{r}}\omega \bar{\mathbf{r}}$$

$$H_o = \bar{I}\omega + m \bar{r}^2\omega = (\bar{I} + m \bar{r}^2)\omega$$

$$H_o = I_o\omega$$



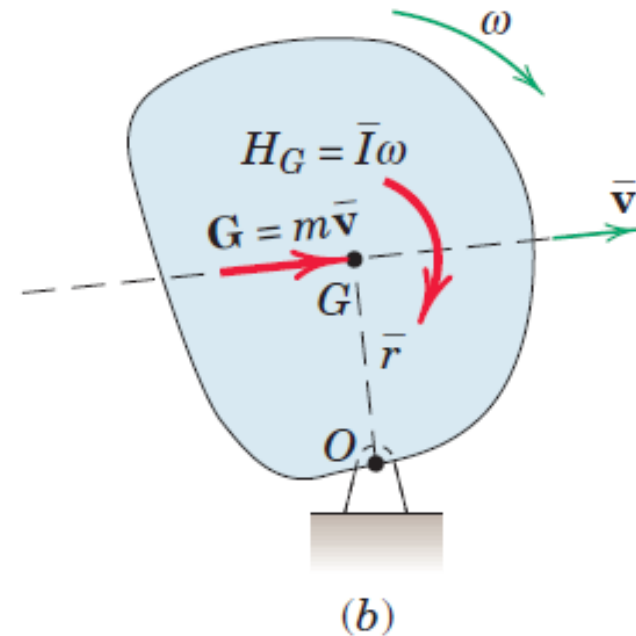
Angular Momentum

$$H_o = I_o \omega$$

$$\sum M_o = \dot{H}_o$$

$$H_{o_1} + \int_{t_1}^{t_2} \sum_o M_o dt = H_{o_2}$$

- Do not add linear momentum and angular momentum for the same reason that force and moment cannot be added directly



Inter-Connected Rigid Bodies

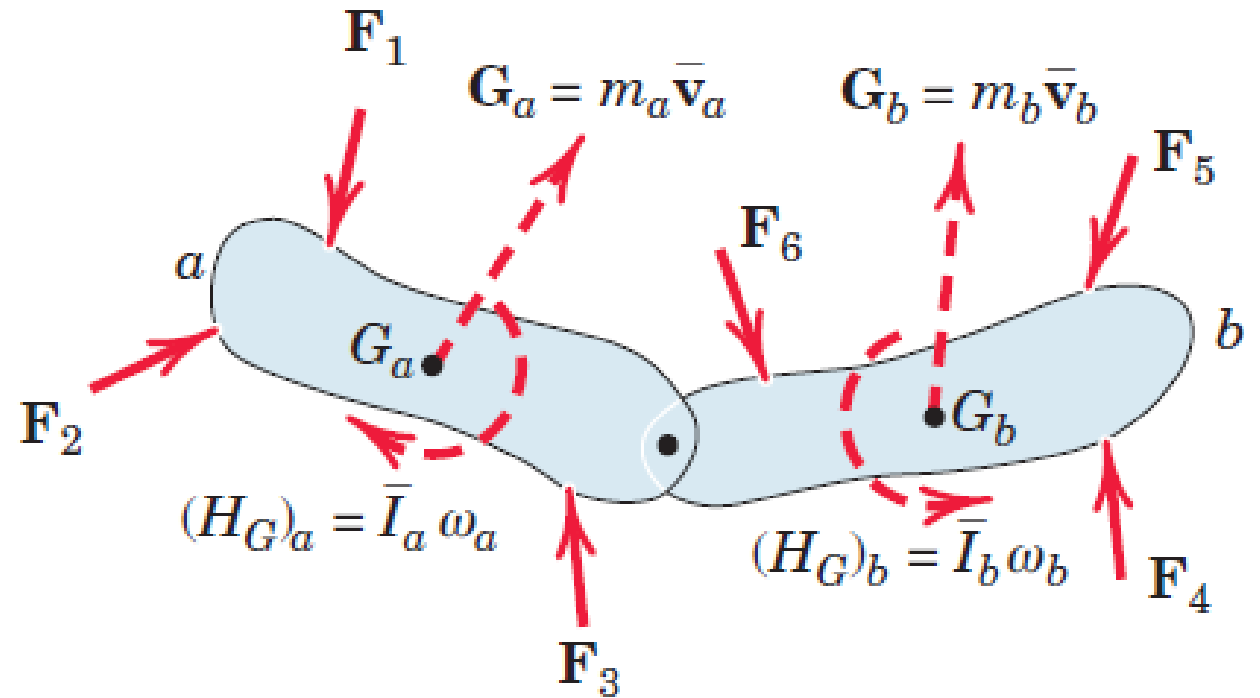
$$\sum F = \dot{G}$$

$$\sum M_o = \dot{H}_o$$

Where O is a fixed reference point

$$\sum F = \dot{G}_a + \dot{G}_b + \dots$$

$$\sum M_o = \dot{H}_{o_a} + \dot{H}_{o_b} + \dots$$



$$H_{o_1} + \int_{t_1}^{t_2} M_o dt = \overline{H_{o_2}}_O$$

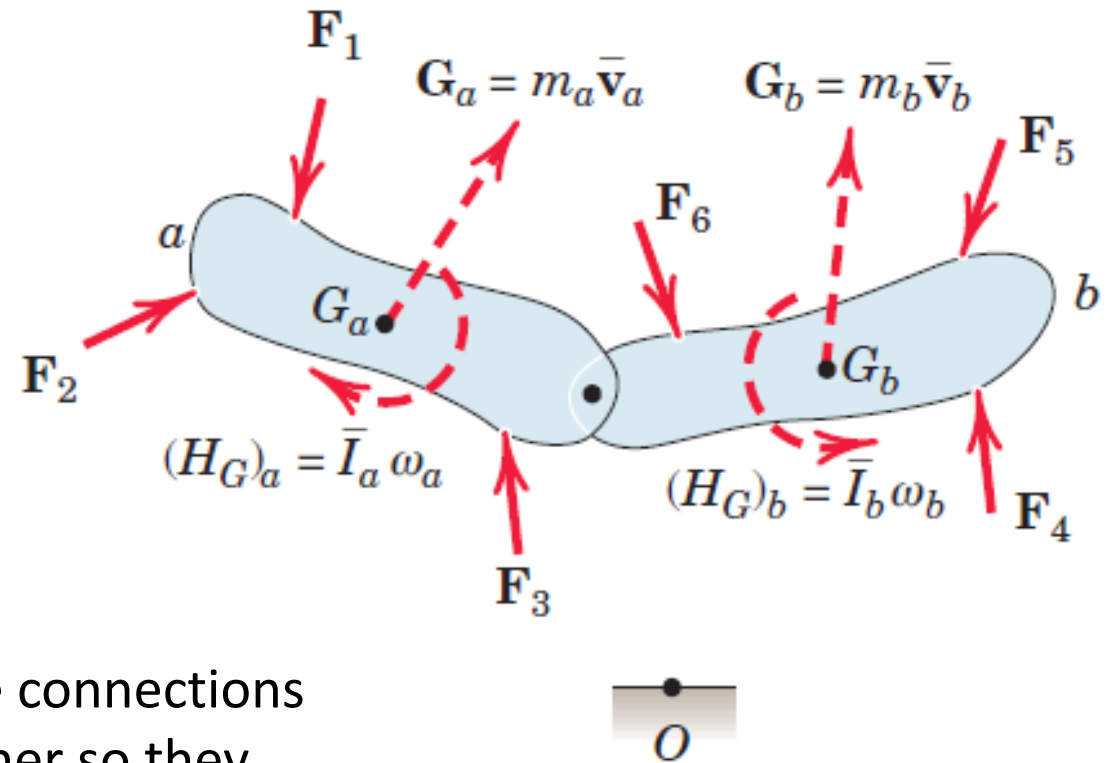
Inter-Connected Rigid Bodies

$$\sum F = \dot{G}_a + \dot{G}_b + \dots$$

$$\sum M_O = \dot{H}_{Oa} + \dot{H}_{Ob} + \dots$$

$$\int_{t_1}^{t_2} \sum F \, dt = \Delta G_{\text{system}}$$

$$\int_{t_1}^{t_2} \sum M_O \, dt = \Delta H_{O\text{system}}$$



- Equal and opposite actions and reactions in the connections are internal to the system and cancel one another so they are not involved in the force and moment summations.
- Point O is one fixed reference point for the entire system.

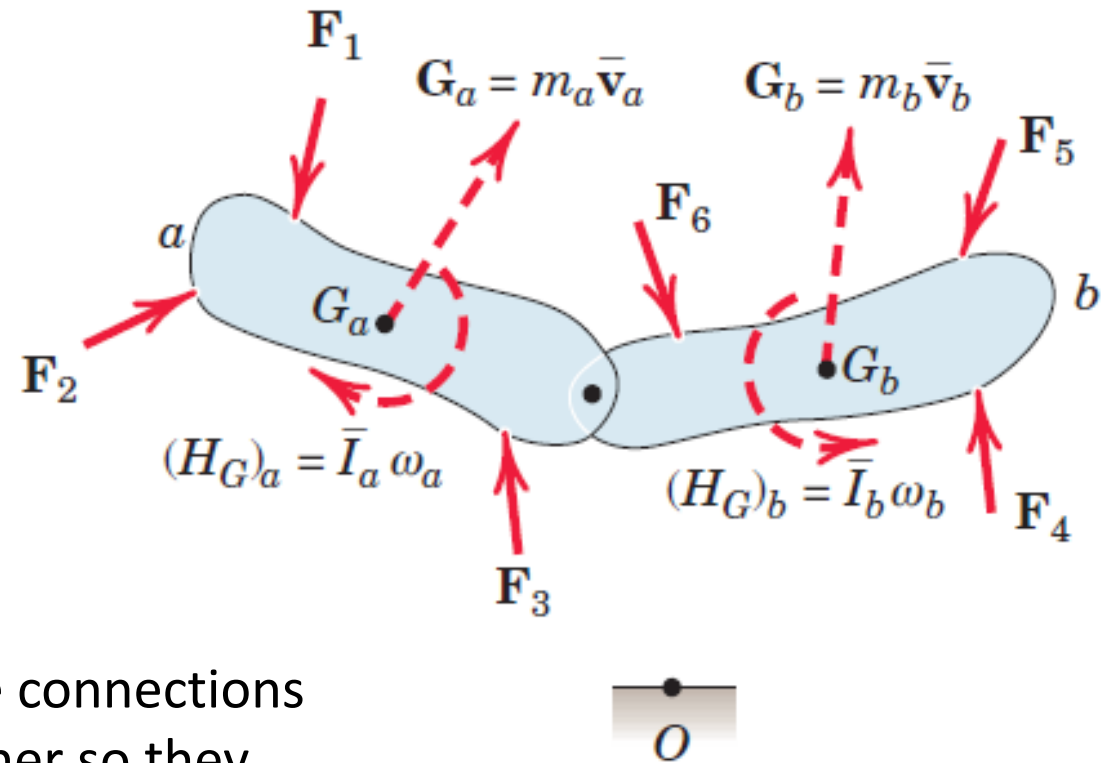
Conservation of Linear Momentum

$$\sum F = \dot{G}_a + \dot{G}_b + \dots$$

$$\sum M_O = \dot{H}_{Oa} + \dot{H}_{Ob} + \dots$$

$$\int_{t_1}^{t_2} \sum F \, dt = \Delta G_{\text{system}}$$

$$\int_{t_1}^{t_2} \sum M_O \, dt = \Delta H_{O\text{system}}$$



- Equal and opposite actions and reactions in the connections are internal to the system and cancel one another so they are not involved in the force and moment summations.
- Point O is one fixed reference point for the entire system.

Conservation of Angular Momentum

- If the resultant moment about a given fixed point O or about the mass center is zero during a particular interval of time for a single rigid body or for a system of interconnected rigid bodies,

$$H_{O_1} = H_{O_2}$$

$$H_{G_1} = H_{G_2}$$

- Angular momentum either about the fixed point or about the mass center undergoes no change in the absence of a corresponding resultant angular impulse

Conservation of Angular Momentum

- In the case of the interconnected system, there may be angular-momentum changes of individual components during the interval, but there will be no resultant angular momentum change for the system as a whole if there is no resultant angular impulse about the fixed point or the mass center

THANK YOU